

Exact Inference, Measurement, and Adversarial Evaluation in Six-Player Canadian Fish

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Abstract

Six-player Canadian Fish is a team card game in which every action is public, so the entire hidden state is the initial deal and the posterior over it is exactly computable. That combination — tractable exact inference, three-player teams that cannot communicate, and a deterministic published policy whose coordination is therefore common knowledge — makes it an unusually clean laboratory for separating what a policy knows from what it does. We know of no prior published computational work on the six-player game.

This paper is the complete account of a six-version programme in that domain. It reports an exactly computable observer-conditioned deal posterior, and the finding that substituting it for the deployed approximation makes the policy *worse*, because exactness here is bought by discarding behaviour; a determinized search that is 35.69 points worse than its own blueprint under an unguarded argmax, and recovered entirely by a paired lower-confidence-bound guard; and a measurement crisis in which five mechanisms cleared a 95% paired interval at one budget and returned exactly zero at three times it. Roughly half the contribution is the apparatus built in response: a duplicate-block design, deal-clustered intervals, power computed before a cell is run, replication across disjoint banks as a veto over pooled significance, sealed holdout material with committed digests, preregistration with stated failure conditions, a commit gate that runs before any strength number, calibrated detection floors, and a mechanical homogeneity gate with planted positive controls.

The programme’s final cycle put one frozen configuration through that apparatus on material sealed before the configuration existed. The preregistered primary claim passes and is certified: against the cheapest point of the previous frontier the configuration scores +3.33 points [+2.88, +3.78] over 48,000 games, replicating on both banks. Eight independent adversary searches, including a white-box inverter that reads the target’s own deterministic policy, all lose. The advantage is not a self-play convention: a single seat’s upgrade is worth +1.93 of the +4.49-point three-seat gain, and three independently-trained runs cross-play within 0.01 points of their own diagonal.

And it is not an advance on what the programme already had. The configuration does not beat the composite assembled two phases earlier — +0.15 points [−0.29, +0.59] — so the increment the intervening architecture work contributed did not replicate. On a 31-member panel scored identically for four arms it is *third of four* on minimax regret, and its own headline worst cell does not replicate in sign. The one quantity falling short of its advance-stated training value is the group of keys the freeze was *selected over*, by 0.63 points against an expected maximum under the null of 0.74.

The residue is therefore negative and, we argue, more durable than the release: four candidate architectures killed with measurements rather than arguments, including a declaration channel that closes at exactly zero and a leaf evaluator fitted 74× better that buys nothing; a corrections register against this project’s own record; and a named barrier — this policy class converts

mechanism into win rate at a rate below a detection floor that does not fall as evaluation games increase. We state what evidence would be required before any Fish-playing team could credibly be called near-optimal, and none of it has been produced.

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1 Introduction

This paper is the account of record for a research programme in six-player Canadian Fish, a team card game with a property that is rare and useful: *every action is public*. Cards move only through asks whose target and outcome everyone observes, so the entire hidden state is the initial deal, and the posterior over it is exactly computable in closed form. A domain in which inference can be made provably correct is a domain in which every remaining loss is attributable to policy — and that separation is what the programme has spent six versions exploiting.

§2 states the game completely, so that nothing here depends on outside material, and §3 says why it is worth studying. The short version is three properties in combination. The hidden state is a single object under hard logical constraints, so exact inference is cheap rather than aspirational. The unit of play is a three-player team whose members cannot communicate, which is the decentralised control problem in its purest card-game form. And a team of three *identical, deterministic, published* agents inverts the standard assumption of the team-correlation literature, in which the correlation device is private: here it is common knowledge, and an adversary may read it.

What this paper contains. The programme has produced two kinds of result, and this paper treats them as equally important.

The first kind is about the game. An exactly computable observer-conditioned deal posterior (§5.3) — and the finding that substituting it for the deployed approximation makes the policy *worse*, because exactness in this game is purchased by discarding behaviour (§6.3.2). More than half of ask decisions ending in a bit-for-bit tie that the *exact* posterior also cannot separate, making the largest item on the design register a property of the information set rather than a defect (§6.3.1). A determinized search that is 35.69 points *worse* than the blueprint it searches from when its action is chosen by an unguarded argmax, and fully recovered by a paired lower-confidence-bound guard (§6.3.3).

The second kind is about measurement, and it is roughly half the contribution. §7 presents the apparatus as method rather than housekeeping: the duplicate-block design, deal-clustered intervals, power computed before a cell is run, replication across disjoint banks used as a veto over pooled significance, sealed material with committed digests, preregistration with stated failure conditions, a commit gate that runs before any strength number, calibrated detection floors, and a mechanical homogeneity gate with planted positive controls. Every one of those components was built in response to a specific measured failure, and most of the failures were this project’s own (§6.3.4).

The final cycle, and its verdict. §§9–13 report a cycle designed so that its answer could not be argued into being positive: characterise the instrument, search for adversaries in the open, build candidate architectures, freeze exactly one configuration, and evaluate it from a cleared context against a protocol committed before any holdout deal had been played.

The preregistered primary claim passes and is certified: +3.33 points [+2.88, +3.78] against the cheapest point of the previous frontier, over 48,000 games of sealed material, replicating on both banks and gate-clean. Nothing built to exploit it succeeds; it is not a self-play convention.

And it does not beat the composite the programme had already assembled two phases earlier — +0.15 points [−0.29, +0.59]. The protocol stated in advance what that would mean, and §15 says it: the strength gain in this cycle is real, it was found by an unfitted coordinate sweep, and the architecture work that followed added no measurable strength on top of it. Two further results point the same way. On a 31-member panel the frozen configuration is *third of four*

on minimax regret. And the one quantity that falls short of its advance-stated training value is the group of keys the freeze was *selected over*.

Contributions.

1. **A complete, self-contained treatment of a domain with no prior computational literature** (§§2–5): rules, dialect, the exact posterior and its closed form, and the verification suite that establishes the engine is what it claims to be.
2. **An evaluation apparatus presented as adoptable method** (§7), each component paired with the failure it prevents — including two occasions on which this project’s own published claims turned out to be artefacts of its instrument.
3. **An operational definition of team exploitability** for a homogeneous team of published deterministic agents (§8), made mechanical by four tests with three calibrated positive controls.
4. **Four candidate architectures killed with measurements** (§10): a declaration channel that closes at exactly zero, a leaf evaluator fitted $74\times$ better that buys nothing, per-decision objectives that produce worse policies while moving their own proxies the right way, and a learned re-ranker worth -0.01 ± 0.45 against a free hash tie-break.
5. **A preregistered evaluation with a stated failure condition** (§12), of which two conditions came out against the release and are reported as such (§15).
6. **A corrections register** (§16) against both the corpus’s record and this cycle’s own, and **an explicit statement of what a near-optimality claim would require** (§18), none of which has been produced.

What this paper does not claim. It does not claim the release is stronger than the best configuration available before the architecture work began; the cell that would show that contains zero. It does not claim the frozen configuration is hard to exploit — only that eight searches, at a stated budget, in stated classes, did not exploit it, which is a lower bound within a searched class and nothing more (§17). It does not claim the gain is located: the protocol’s own location test does not replicate across banks (§15.2). And it does not claim that any policy here is near-optimal, or that this is the strongest Fish-playing team, because the comparison class has one member and every exploitability number in this corpus is a lower bound from a search rather than a bound on what a search could find.

2 Six-player Canadian Fish

This section states the game completely, so that nothing later in the paper depends on outside material. Readers who know the game should still read §2.2: published descriptions differ on several consequential points, and the dialect studied here is one explicit selection among them.

2.1 The rules

Canadian Fish — also called Literature — is played by six players in two teams of three, seated alternately so that no two teammates are adjacent. The deck is 54 cards: the standard 52 plus two jokers. It is partitioned into nine **half-suits** of six cards each — for each of the four suits, the low half 2–7 and the high half 9–A — plus a ninth half-suit made of the four eights and the two jokers. Every card belongs to exactly one half-suit, and each player is dealt nine cards. The team that takes at least five of the nine half-suits wins.

Play proceeds by **asks**. On their turn a player names one opponent and one card. The ask is legal only if

1. the named card lies in a half-suit that has not yet been claimed;
2. the asker holds at least one *other* card of that half-suit;
3. the asker does *not* hold the named card; and
4. the target is an opponent who still holds at least one card.

If the target holds the card they must surrender it, and the asker keeps the turn. If they do not, the turn passes to the target. Condition (2) is what makes the game a game: a player may only ask about a half-suit they already have a stake in, so every ask is a public declaration of partial ownership.

A half-suit is claimed by a **declaration**. A declaring player names, for each of the six cards of a half-suit, which member of their *own* team holds it. The claim is all-or-nothing: if every one of the six assignments is correct the declaring team scores the half-suit, and if any assignment is wrong the opposing team scores it. A player may not declare a half-suit they hold entirely — the interesting case is always a claim about cards in teammates’ hands.

Everything is public. Every ask, its answer, every card transfer, every declaration and its outcome, and every player’s card count are observed by all six players. Nothing is hidden except the initial deal.

2.2 The dialect studied, and why the choice is explicit

Published descriptions of this game document mutually incompatible variants [1], so the rule set below is a selection rather than a canon. It matches the strategy corpus this line of work draws on [2] and the rules of record supplied by the project owner. Every contested choice is an engine switch, so its effect is measurable rather than assumed, and §13.6 measures the headline result under 8 of them.

The consequential choices are:

- a 54-card deck in nine half-suits, nine cards per player;
- declaration legal at any moment, in or out of turn, by a seat holding cards or not;
- a misdeclaration awarding the half-suit to the opposing team;
- lowest seat index first among simultaneous declarers;
- an eight-level willingness ladder in the **forced endgame**, which arises when one whole team is cardless and the other must continue declaring;
- a 400-ask action cap, whose residue is adjudicated neutrally by physical majority.

No prearranged signalling conventions are available to any policy. Team communication is limited to the forced-endgame willingness bit and to the successor a cardless turn-holder names. Appendix A lists each choice against the switch that changes it.

2.3 Two places the engine is deliberately more restrictive than the rules allow

Both are coordination channels a policy could in principle exploit, and both were measured rather than argued about. They are stated here because a reader entitled to ask “why not let the team talk about it?” deserves the number rather than the principle.

Turn transfer. When the turn reaches a cardless seat whose teammates both still hold cards, the rules of record permit the team to negotiate openly over who receives it, sharing nothing but willingness. The engine instead has the cardless seat choose unilaterally from its own belief. The channel is empty: a genuine multi-candidate transfer decision arises 0.148 times per game, and replacing the unilateral rule with a *ground-truth* chooser on one team, paired on common deals, is worth -0.0007 ± 0.0024 half-suits per game over 6,000 paired games, with the two arms diverging at

all in 0.48% of games. A legal mechanism cannot beat the clairvoyant one, so that interval bounds the entire channel.

Declaration arbitration. The rules do not say who acts first when two seats offer a declaration at the same instant, so the order is a modelling choice. The engine scans by lowest seat index, deliberately declining to rank candidates by stated confidence, because a confidence comparison moves a statistic of one seat’s hand to five others. That refusal costs 0.37 pp of win rate pooled over 30,000 games (range +0.28 to +0.45 pp across five seeds), and a clairvoyant arbitrator — one that picks a proposer whose named allocation is actually correct — is worth +0.35 pp on the same design. The information-safe rule forgoes about a third of a percentage point, and no rule of any kind, safe or unsafe, informed or omniscient, recovers appreciably more.

3 Why this game is worth studying

Fish sits at an unusual point in the space of imperfect-information games, and the point is unusual in a way that makes certain questions cheap to ask that are expensive everywhere else. This section says what those questions are, because they are the reason the rest of the paper exists.

3.1 The hidden state is the deal, and the posterior over it is exact

Every card movement in Fish is public. A card changes hands only through a successful ask, and the ask, its target and its outcome are seen by everyone; a declaration reveals an entire half-suit’s allocation whether it succeeds or fails. It follows that the *only* thing any player does not know is the initial deal, and that the public record determines every subsequent card position given that deal.

This is a stronger property than it sounds. In most hidden-hand card games the reachable-history set is what must be filtered, and history filtering is FNP-complete in general [23]. Fish escapes: there is exactly one object to reconstruct, and the constraints on it are logical rather than probabilistic. A player who has been told “I do not hold the seven of hearts” knows it, forever, with certainty. The observation history is a conjunction of hard constraints on a fixed unknown, and the posterior over deals consistent with those constraints is computable in closed form (§5.3).

The consequence for research is that Fish lets you *separate* inference from policy. In poker or Bridge, a claim that a policy is weak can always be answered with “its beliefs are wrong”. Here the beliefs can be made provably right, at which point every remaining loss is attributable to the policy. That separation is what makes the negative results of §6 and §10 meaningful rather than merely discouraging.

It also sets up the single most surprising finding in this corpus, which §6.3.2 reports: making the belief exact makes the policy *worse*.

3.2 The team is the unit, and the team cannot talk

Three teammates share a payoff, hold different private information, and have no communication channel beyond the public actions themselves. That is precisely the decentralised control problem the common-information approach formalises: reformulate from the point of view of a coordinator who knows the common information and issues *prescriptions*, maps from each controller’s private information to its action [52]. In Fish the common information is the entire public history, so the coordinator’s information state is exactly the belief object the engine already maintains.

Two things follow that matter for evaluation rather than for algorithms. First, the coordinator’s problem is not solvable here — a prescription maps every possible nine-card hand to an ask — so the reduction is a correctness lens and not a method. Second, and more usefully, *any deterministic procedure that reads only public state and its own hand is automatically common knowledge among the three teammates*, with no shared seed and no communication. That is a cheap version of a property the cooperative-search literature has to work for [28], and it is why the side-channel definition of §8 can be made mechanical.

The qualifier is load-bearing, and §6.3.1 is where it bites: a randomised tie-break loses the property unless its randomisation is seeded from the public history.

3.3 A team of published, deterministic, identical agents inverts a standard assumption

The team-correlation literature uniformly assumes the correlation device is the team’s own — private, and unknown to the opponent [53, 54, 55]. A homogeneous team of three *identical published deterministic* agents inverts that: the device is common knowledge, and an adversary may read it. The threat model of §8 takes that seriously, and §14 reports what an adversary that reads the target’s own policy and inverts its transcript actually achieves.

We found no prior work covering this case, and §4 states the scope of that search rather than claiming priority beyond it.

3.4 The domain is cheap, and that is a methodological asset

A Fish deal plays out in roughly a hundred public events. The engine used here plays hundreds of complete six-player games per second per core with exact inference running in every seat. This is not a boast about engineering; it is what makes the evaluation programme in §7 possible at all. A study that needs 48,000 paired games to resolve a three-point effect, two disjoint holdout banks for every claim, a 31-member opponent panel scored identically for four arms, and eight independent adversary searches, can run all of that on one desktop in a day. In a domain where a single evaluation cell costs GPU-hours, the honest version of this paper could not have been written.

The corresponding liability is stated in §17: cheap evaluation is not external evaluation, and every number here was produced by one engine that this project wrote.

3.5 What the game does *not* offer

Two disclaimers, because both bear on how the results should be read.

Fish’s perfect-information game is degenerate, not merely fused. A clairvoyant player names only cards the target actually holds, so never fails an ask, never loses the turn, and declares each half-suit as soon as their team holds all of it. Almost every reachable position carries the same perfect-information value. Determinization — the standard recipe for hidden-hand card games [11, 12] — therefore collapses here to choosing the ask most likely to hit, blind to the information an ask leaks, the information a refusal supplies, and the option value of postponing a declaration. §4.2 places this against the literature, and §6.3.3 reports what the project measured when it built such a search anyway.

And Fish has no passive continuation. A player must ask or declare; there is no equivalent of poker’s check/call rollout, which is what the standard cheap exploitability probe is built on [38]. §18 names this as the single largest obstacle between this corpus and a defensible near-optimality claim.

4 Related work

This section covers the work this study builds on and the classes of work it deliberately declines. Absence is a design decision here, not an omission, and it is recorded as one.

4.1 Fish and Literature

The published material on this game is human and informal, and neither of its two substantial sources is quantitative or describes a program. McLeod’s Pagat entry [1] is the canonical rules source and records several points on which dialects differ; ours is specified in §2.2. Develin’s chapter [2] is the most detailed strategy writing we located, describes the same dialect, and supplies two observations this design uses: an ask teaches the opponents as much as it teaches a teammate, and a player may not declare a half-suit they hold entirely. Secondary strategy pages [3, 4, 5] restate blackballing, asking the asker, and an endgame delegation heuristic; they are evidence about how the game is played and talked about, not sources of quantitative claims.

Computational prior art remains close to absent. Searches across this project’s cycles found no academic work on this game, no public repository containing search, planning or a working learner for it, and no coverage in RLCard [9]. The one agent located, Somani’s *literature* [6, 7], plays the four-player, 48-card variant, which does not raise the six-player allocation problem, and publishes no strength numbers; a closed-source mobile application advertises bots with no stated methodology [8]. Every novelty statement in this paper therefore takes the form “we found no prior published work that ...” and should be read against that scope. One recent item is worth naming because it may become the external calibration this line of work lacks: Valet packages twenty-one traditional imperfect-information card games in a common description language with measured branching factors and durations [10], though whether any of the twenty-one is a team game is not stated on its abstract page and we did not fetch the full text, so nothing here is compared against it.

4.2 Determinization, and why double-dummy analysis fails here

Determinization, or perfect-information Monte Carlo — sample hidden cards consistently with the public history, solve the resulting perfect-information game, and play the move with the best average value — is the standard recipe for hidden-hand card games [11, 12], and is unsound however many worlds are drawn. Frank and Basin named the two mechanisms: *strategy fusion*, in which the searcher plays a different strategy in each sampled world although a real player must commit to one strategy per information set, and *non-locality*, in which a node’s value depends on parts of the tree outside its own subtree [13, 14]. Long, Sturtevant, Buro and Furtak characterise when the method nevertheless performs well, through leaf correlation, bias and a disambiguation factor [15]; Information Set Monte Carlo Tree Search replaces independent trees with a single tree over information sets [16, 17]; the $\alpha\mu$ family backs up Pareto fronts of world-vectors instead of scalar averages, repairing both mechanisms, and reports Bridge 3NT declarer play at 60.2% to 63.0% with twenty worlds [18]; and knowledge-based paranoia search reasons about the worst case over deduced-consistent worlds rather than the average, reporting over 1,000 points on the extended Seeger scale in Skat [19].

Fish breaks the method for a reason none of that literature addresses, and §3.5 states it: the perfect-information game is degenerate rather than merely fused, so $\alpha\mu$ ’s repair of unsoundness does not rescue the construction.

What this corpus adds is that the binding problem is not the abstract failure but *the statistic*. One determinization’s value is the final half-suit differential, whose spread is an order of magnitude

larger than the differences between the leading candidates, so an argmax over sampled means selects on residual noise — and does so more confidently the more candidates it is offered. That is the optimiser’s curse rather than strategy fusion, and §6.3.3 reports it measured at 35.69 points by a control that forces the same code to return the blueprint’s own pick. The transferable claim from that result is the guard, not the search.

4.3 Search in imperfect-information and cooperative games

States in an imperfect-information game do not have well-defined values, so depth-limited search needs something at the leaf that represents the opponent’s ability to choose. Brown, Sandholm and Amos supply it: let the opponent select among k continuation strategies at the depth limit, each yielding its own leaf-value vector, and require the searcher to be good against that choice [24]. The soundness argument is stated for two-player zero-sum games, and Fish’s team-level game presents three uncorrelated opponent seats at the leaf, so nothing is inherited; it is adopted here as a robustification heuristic with a measurable consequence, and audited rather than assumed.

Two lines address what a search may prune. Zhang and Sandholm solve subgames over low-order knowledge rather than the full common-knowledge closure, and prove that the naive form can *increase* exploitability [25]. That warning applies directly to the live-ask gate this policy family ships, which is a knowledge-limited pruning rule, and it is why the side-channel and pathology gates of §7.6 were built before any search work rather than after it. Subgame solving in adversarial team games builds a gadget over the team belief DAG and runs in polynomial time given a best-response oracle *when the blueprint is sparse* [26]; a Fish blueprint is not sparse, so that escape hatch does not open here.

In cooperative games the constraint transfers verbatim. SPARTA improves a blueprint purely at test time and distinguishes single-agent search — which assumes every other agent strictly follows the blueprint — from multi-agent search, in which all agents run the *same* common-knowledge procedure [28]. The moment a second teammate also searches, the first teammate’s belief over its partner is computed under a policy that partner is no longer playing, and the improvement guarantee is void. Fish satisfies the requirement cheaply, for the reason §3.2 gives; the qualifier about randomisation is what §6.3.1 turns into an engine switch.

The full-solver line [37, 35, 36] established that test-time re-solving over public belief states is principled rather than a hack. The algorithms are declined here, because each requires iterating or learning over a closure whose cardinality at deal time is $45!/(9!)^5 \approx 1.9 \times 10^{28}$ deals from one seat. Update equivalence recasts search as replicating a last-iterate-convergent update locally, needing no public-information enumeration [27]; its improvement guarantee is for common-payoff games, which Fish is not. Learned Belief Search [29] is redundant here, because this game’s exact belief is already computed and oracle-validated. Online planning by policy fine-tuning [30] requires gradient updates at decision time and does not transfer to a CPU-only engine with no network. The depth-limited counter-strategy line [31, 32] carries one warning this study honours — robust-response methods do not compose with limited-look-ahead solving off the shelf — and the learned-model and gadget-refinement work [33, 34] is recorded for its meta-lesson rather than its machinery: when a search subproblem has many near-equivalent optima, which one is selected is not a tie-break. That is the same observation §6.3.1 makes about an argmax resolved by array order.

Every method above is published with a caveat that it can increase exploitability, so none can be evaluated by win rate alone. Local best response gives a cheap lower bound: a small greedy search, at most two actions deep, showed every entrant in a 2016 poker competition to be more than 3,180 mBB/h exploitable [38]. Its two documented pitfalls are carried into §14 — restricting where the response may begin understates exploitability, and a negative score proves nothing

— and every exploitability figure in this paper is a lower bound within its response class.

4.4 Policy-aware inference

Skat engines enumerate the deals consistent with the public record and reweight them by a model of what each player would have done — a supervised per-card location model [21], or reach probabilities taken from a policy [22] — and a dedicated inference module was reported worth +217 tournament points per 36 games to the engine Kermit on its own [20]. The policy prior of §5.4 is the same idea in a much simpler form: a fitted soft reweighting of the rules posterior, with two parameters and no learned opponent model.

This is the only literature that has measured the same trade-off, and it is where §6.3.2’s central negative result belongs. What is new there is the margin and its direction: in this game the exact conditional posterior is computable in closed form and cheap, so the usual excuse for approximating does not apply — and yet substituting the exact posterior into an otherwise identical policy *loses*, on two independent seed banks.

Two adjacent lines are named and not used. Belief modelling in cooperative games — the Bayesian Action Decoder’s public-belief formulation [40] and its factorised-prescription successor [44] — is aimed at games whose exact belief is intractable, which is the opposite of the situation here. And the exact-counting machinery this study does not need to approximate — Ryser’s inclusion–exclusion formula [58], Glynn’s sign-vector formula [59], the Jerrum–Sinclair–Vigoda chain [60], matrix scaling [61] and sequential importance sampling [62] — is sidestepped entirely by the block structure of §5.3. No learned belief model appears anywhere in this corpus, and that is a stated non-goal rather than an oversight.

4.5 Team games and common information

The formal statement of three teammates sharing a payoff, holding different information and unable to communicate is the common-information approach [52], discussed in §3.2. The team-game equilibrium literature is cited to delimit scope: the target concept for a Fish team is a team-maxmin equilibrium with correlation [53], connected to extensive-form correlation by Farina and coauthors [54] and made computable through the team belief DAG and the two-player conversion [55, 56], with subsequent work on hidden-role settings [57]. None of it is computed here. The deeper reason a Fish team resists the DAG is that its meta-player fails A-loss recall, for exactly one reason: each seat privately observes nine cards its teammates do not.

The cooperative-play line is cited for its evaluation discipline rather than its algorithms. Hanabi was proposed as a benchmark precisely because self-play scores overstate coordination [39], and the ad-hoc-teamplay results that followed [41, 42, 43, 45, 46, 47, 48, 49, 51] are the reason the cross-play measurement of §13.5 exists at all: a convention that collapses when partners are swapped is brittleness, and it is invisible to self-play evaluation. The same lesson from a different direction is Cicero’s use of a human-regularised policy [50].

4.6 Evaluation methodology

The apparatus of §7 draws on a literature that is usually treated as plumbing. Duplicate-style variance reduction and its estimators — DIVAT [64] and AIVAT [65] — are the reason every cell in this paper is a paired duplicate rather than an independent sample. The Annual Computer Poker Competition’s protocols [63] are the model for a fixed opponent panel scored identically for every arm. Abstraction pathology [66] is the standing warning that a model can be improved on its own metric and made worse in play — which §10.2 reproduces exactly. Re-evaluation of agent

populations [67] and the rating literature [68, 69, 70] inform why this paper reports a worst case and a minimax regret over a shared panel rather than a single rating number. Recent work on gold-standard evaluation practice [71] is consistent with the preregistration and sealing discipline of §7.5, which this project adopted independently.

5 The engine, and exact inference over deals

Everything in this paper is measured by one program, and half the paper’s contribution is the discipline that program enforces. This section describes it: the simulator, the information barrier between the rules and the policies, the exact deal posterior, and the verification that all three are what they claim to be.

5.1 The simulator

The engine is a single-header C++20 core. A hand is one 64-bit word over the 54 card indices $c = 6h + i$ for half-suit h and offset i , so hand arithmetic is bit arithmetic. Random streams are counter-based and keyed by *(seed, seat, purpose)*, which is the property everything else rests on: **a deal and its entire play reproduce from the seed alone**. The duplicate-block protocol of §7.1 depends on that and on nothing else.

Throughput on the deployed policy is 303.4 complete six-player games per second, against 276.3 for its predecessor, with exact inference running in every seat. Every compute figure in this paper is single-machine, measured under load, and therefore a lower bound.

5.2 Information safety

The acting policy never receives the game state. It receives its own `Knowledge` object — the constraint bookkeeping of §5.3 — plus its own hand and the public event stream. There is no pointer from a policy to the deal, to another seat’s hand, or to the referee. This is an architectural barrier rather than a convention, and it is what makes the side-channel audit of §7.8 a test of the *policy* rather than of the harness: a policy that cannot reach the deal cannot cheat by reading it, so any decision the audit fails to reproduce from public information indicates state the policy carried across deals, not information it stole from the referee.

5.3 The hidden state is the initial deal

Because every card movement is public (§3.1), an observer’s uncertainty is entirely about the initial deal, and the observer’s information is a conjunction of hard constraints on it:

- (C1) its own nine cards, known exactly;
- (C2) each other seat’s current card count, known exactly;
- (C3) every card that has been transferred, and to whom;
- (C4) for each failed ask, that the target did not hold the named card at that moment;
- (C5) for each ask, that the asker held at least one other card of the named half-suit and did not hold the named card.

The counting problem this induces would in general be a permanent-like enumeration over the whole deal, which is the object the literature approximates with Ryser’s formula [58], Glynn’s [59], the Jerrum–Sinclair–Vigoda chain [60], matrix scaling [61] or sequential importance sampling [62].

Fish sidesteps all of it, because the constraints factor by half-suit. Constraint (C5) is about a half-suit; (C4) is about a card, hence about a half-suit; the only cross-half-suit coupling

is (C2), the per-seat card counts. So the count decomposes into a per-half-suit block enumeration joined by a small dynamic program over the count vector. The result is exact in closed form: Z , the per-card marginals, the probability a team owns a half-suit, and the probability of a named allocation, all without sampling. An exact rebuild costs 0.4 ms, which is why this project has been able to treat inference as solved and spend its cycles on policy.

5.4 The policy prior, and the object the policy actually deploys

The exact posterior is exact *under a uniform prior over deals consistent with the constraints*. It contains no model of behaviour: the block construction consumes the deduction state alone and has no parameter through which an opponent’s tendencies could enter.

The deployed path adds one: a fitted soft reweighting of the rules posterior with two parameters and no learned opponent model, in the spirit of the Skat inference line [21, 22, 20] but far simpler. The shipped weight is $\theta = 0.37062$.

This is where the corpus’s most-cited internal result lives, and §6.3.2 states it: scored as predictors on identical states, the *exact* posterior is the worst of the three inference paths measured. It is the reason this project’s standing methodological rule is that “exact Bayesian inference” is an assumption to test rather than a property to assume — a rule that paid off a second time in §10.1.

5.5 Verification

Three independent checks run over the engine, and all three ran again as preconditions of the evaluation in §13.

Soundness audit. Every seat’s deduced knowledge is checked against the true deal at every decision: no agent’s deduction may ever exclude the truth. The evaluation reported here recorded 0 violations in 6,737,436 checks, and the preceding cycle 0 in 23,594,580.

Cross-engine agreement. The block dynamic program, an independent card-counting path and an exact sampler are required to agree. The self-test reports 424,986 checks passing, with block-versus-card-DP agreement at the limit of double precision.

Identity control. The current policy, carrying its predecessor’s parameter vector with every new mechanism switched off, must produce a transcript identical to its predecessor’s under md5. It does (PASS). This is what makes an ablation an ablation: without it, a measured difference could be an accident of refactoring.

Two engine defects found by these checks are worth naming here, because both changed previously published numbers and both are the kind of defect that survives precisely because it is invisible to the deployed path.

The exact reference engine was unsound in multi-agent play. The block dynamic program parked every instance’s tables in a shared per-thread pool, so any second construction on the same thread silently repointed the first instance’s tables: 285 mismatches in 294 checks. Every previously published number computed with the exact belief in a configuration where more than one seat holds such an object is affected. The deployed approximate-belief numbers are not — which is exactly why the defect survived three studies. It is repaired at the query level; raw reads of the shared pool still differ (175) and are documented rather than fixed, because the pool is what keeps the memory footprint of six concurrent observers bounded.

A declaration pre-gate audit was dead code. The audit that verifies the declaration pre-gate never rejects a *correct* declaration was parsed only in an older branch and so had never run. Once

enabled it reports 0 false negatives over 5,472,906 declaration opportunities and 14,449,770 gate rejections: the pre-gates are exact.

6 The research arc

This project has run for six versions. The arc matters to the reader of this paper for one reason: the v0.7 evaluation of §13 is only interpretable against what came before it, and what came before it includes two occasions on which this project’s own published claims turned out to be artefacts of its measurement rather than properties of its policies. A reader who wants to know why the apparatus of §7 is as heavy as it is will find the answer here.

6.1 v0.2 to v0.4: building the object

The first versions established the domain and the inference. A count-conditioned, empirically tuned linear policy in a TypeScript simulator became a rules-exact C++ engine with an exact reference posterior, a fast deployed approximation, a twenty-feature ask score and a value-based declaration rule. The substantive contribution of that period is the object described in §5.3: an exactly computable observer-conditioned deal posterior in a game where nobody had computed one, cheap enough to run in all six seats simultaneously.

That period also set the pattern this paper spends a great deal of effort undoing. Its evaluation was a few thousand games per cell against a handful of opponents, its ablations were unpaired, and its conclusions were drawn from point estimates. §16.1 lists what that cost.

6.2 v0.5: a failure mode found, removed, and honestly reported as worth nothing

v0.5 diagnosed a two-question deadlock in mirror play — two policies alternately asking each other for cards neither would surrender, ending in misdeclaration — and removed it. It then reported, in terms, that removing it was worth almost nothing in win rate.

That report is the honest ancestor of this one. It is also the point at which the project’s central difficulty became visible: a release that removes a real defect and gains no measurable strength is either at the ceiling of its policy class or is measuring badly, and those two possibilities call for opposite responses.

6.3 v0.6: the measurement crisis

The v0.6 cycle set out to find several points of accessible headroom that the accumulated design register described. It found none of it, and in the process discovered that most of the apparatus was broken. Four results from that cycle are load-bearing for this paper.

6.3.1 Exchangeable ties are irreducible, and the tie-break is a coordination decision

At 54.74% of ask decisions two or more candidates score *bit-for-bit identically*, and the winner is whichever the enumerator emitted first. 93.77% of those ties are two cards of one half-suit at one target, and the *exact* posterior — not merely the approximation the policy deploys — assigns them identical probability in 0.00% of cases. Every rule that ranks them by that probability therefore realises the same hit rate, and so does enumeration order.

The largest item on the inherited register was not a defect. It was a property of the information set.

What is *not* irreducible is the dependence on enumeration order itself. An unstable sort order deciding half of all contested decisions is a correctness problem rather than a strength problem, and §3.2’s common-knowledge property says how to fix it: replace the order with a hash of the public event stream, which every teammate can compute and which no seat’s private hand influences. That switch is one of the five keys of the configuration this paper evaluates (§11), and §10.4 reports what happened when a later cycle tried to do better than randomising inside the tie group.

6.3.2 The exact posterior is a worse predictor than the approximation

Scored as predictors on identical states — mean predictive log loss over unresolved cards, and the realised hit rate of each posterior’s own argmax ask — the ranking is:

Posterior	Log loss (nats)	Argmax hit rate
Exact, uniform deal prior	1.42246	47.94%
Rules posterior, no policy prior	1.39083	49.99%
Deployed approximation	1.38218	51.49%

The exact object is the worst row. The policy prior is the entire difference: exactness here is bought by discarding behaviour, because the block construction of §5.3 has no parameter through which behaviour could enter. Exactness and behaviour-awareness are not available in the same object in this game.

This closed an experiment three studies had listed as outstanding, and it established the standing rule this project has applied ever since — “exact Bayesian inference” is an assumption to test. §10.1 is the second time that rule paid, on a different channel.

6.3.3 The optimiser’s curse in determinized search

v0.6 built a determinized search that samples full deals from the public posterior and plays each forward with all six players reconstructed at their own information sets — not clairvoyant, which §3.5 explains would be degenerate.

Chosen by an unguarded argmax over sampled means, that search is **35.69 points worse** than the blueprint it searches from, measured against a control that forces the same code to return the blueprint’s own pick at identical budget, seed and sample size. The reason is the statistic, not the search: one determinization’s return is the final half-suit differential, whose spread is an order of magnitude larger than genuine differences between candidates, so the argmax selects on residual noise and does so more confidently the more candidates it is offered.

A paired lower-confidence-bound deviation rule recovers all of it. Guarded, the search is worth 52.64% against the deployed policy over 2,160 games at 3 banks. The transferable result is the guard rather than the search, and the guarded search — shipped switched off in v0.6 — is one of the five keys of the configuration this paper evaluates.

6.3.4 The evaluation could not resolve a point, and the optimiser could not move a policy

Two apparatus failures, both of which this paper’s methodology exists to prevent.

The evaluation. Pooling every same-pair cell in the v0.6 battery gave 50.53% with a naive z of 1.13 — a null, and an earlier draft of that report published it as one. Re-run at high power across 7 disjoint banks, the same comparison is 50.89% over 126,000 games [50.61, 51.16], above parity on

all 7 banks, worst bank 50.29%. The two cells that had gone the other way returned above parity at the larger sample. The effect was always there; the instrument could not see it.

Five mechanisms in that cycle cleared a 95% paired interval at the per-cell budget this project had used since v0.3 and returned *exactly zero* at three times it. That is the fact that makes every sample size in §7.3 non-negotiable.

The optimiser. The previous fitting harness could not move a policy: one step size for thirty-four coordinates spanning two and a half orders of magnitude, an unpaired objective, and an objective documented as a worst case that was in fact a weighted mean. Its forty-generation refit was indistinguishable from sampling noise, so the vector it “found” was the previous version’s. The repaired optimiser then found a policy that trades nothing head to head for a large replicated gain against the deception archetype the project owner brought from live play (8.60 points).

6.4 What v0.6 handed forward

A policy about nine tenths of a point stronger than its predecessor, a design register most of whose entries had just been closed as worth nothing, and a diagnosis: *at the budget this project had always used, this policy class looks flat*. v0.6 stated plainly that it did not know whether the class *is* flat or whether the instrument was still too coarse.

v0.7 is the cycle built to answer that, and it was organised so the answer could not be argued into being positive. Five phases ran under separate briefs, each starting from a cleared context:

1. **Rebuild the instrument** and characterise what it can and cannot see (§9).
2. **Generate adversaries in the open** and find what actually beats the v0.6 frontier — not the deployed policy, the frontier (§9.3).
3. **Build candidate architectures** against the register’s ranked hypotheses (§10).
4. **Bake off and freeze** exactly one configuration (§11).
5. **Evaluate it** on sealed material under a protocol committed in advance (§12, §13).

The rest of this paper is that programme and its result.

7 Methodology

Half of what this project has produced is the apparatus, and it is presented here as method rather than as housekeeping, because every component of it was built in response to a specific measured failure — most of them this project’s own. A reader working in another game can adopt these pieces individually; each is stated with the failure it prevents.

7.1 The duplicate-block design

What it is. A *cell* is a comparison of two configurations *A* and *B* over a *bank* of deals identified by one integer seed. Every deal in the bank is played twice: once with *A* in the odd seats and *B* in the even seats, and once with the seats swapped. Both configurations therefore play the identical deal from both sides.

What it prevents. Fish has enormous per-deal variance — a deal can be won or lost on the cards — and an unpaired comparison spends most of its sample resolving that variance rather than the difference between the policies. The rotation cancels the deal’s intrinsic luck exactly, in the same spirit as duplicate bridge and the poker line’s variance-reduction estimators [64, 65].

What it requires. Reproducibility from the seed alone (§5.1). If the engine’s randomness is not keyed by (*seed, seat, purpose*), the two rotations are not the same deal and the design silently degrades into an unpaired one.

Throughout this paper a cell’s result is reported as an **edge** in points: $100 \times (\text{winRate}_A - 0.5)$, where 0 means the two configurations are indistinguishable.

7.2 Deal-clustered intervals

The two rotations of one deal are *not* two independent observations. They share a deal, and their outcomes are strongly negatively correlated by construction. Treating them as independent — which is what a Wilson interval on the game count does — understates the width.

Every interval in this paper is a **deal-clustered bootstrap**: 20,000 resamples in which the resampling unit is the *deal*, carrying both of its rotations together. No Wilson interval is quoted anywhere in this paper, and the substitution is not cosmetic: it is the difference between a claim that replicates and one that does not.

Pooling two banks. Two banks of equal size pool as the mean of the two edges with $\text{se}_{\text{pooled}} = \sqrt{\text{se}_1^2 + \text{se}_2^2}/2$, where each se is read back off that bank’s clustered interval. Both per-bank values are printed beside every pooled figure in this paper, so a reader can always see the pooling rather than take it on trust.

Differences of cells. A difference between two *pooled* cells — for example an ablation’s leave-one-out drop — is the difference of the estimates with the two pooled half-widths combined **in quadrature**. This is conservative. The arms are paired on deals, so a genuine paired delta would be narrower, but the harness produces no paired delta *across* two separately run cells, and inventing one would be a stronger claim than the data structure supports.

7.3 Power, computed in advance

At parity a cell’s 95% half-width in points is approximately $98/\sqrt{N}$ for N games. The harness computes this for every cell and emits it with the result, including the mirror case where no half-width exists.

The arithmetic is stated in advance because it decides what may be attempted. Resolving one point needs roughly 9,604 games; a quarter of a point needs sixteen times that. A study that wants to find quarter-point mechanisms on the scoreboard cannot afford to, and §18.1 names that as an open problem rather than pretending it is solved.

The failure this prevents is the one §6.3.4 describes: five mechanisms that cleared a 95% paired interval at one budget and returned exactly zero at three times it. Sample sizes in this paper are chosen from the arithmetic before the cell is run, never after seeing it.

7.4 Replication as a veto

Every claim in this paper is measured on *two disjoint banks*, and the rule is absolute:

A claim whose **sign** does not agree on both banks is reported as NOT REPLICATED, whatever its pooled interval says.

This is stricter than a pooled significance test and deliberately so. A pooled interval can exclude zero while the two banks disagree, and in this corpus that configuration has occurred and has been published as a positive result before. §15.2 is where the rule bites against this paper’s own release, and §13.1 is where it bites against this paper’s own headline cell.

7.5 Sealed material and preregistration

The problem. A programme that generates candidate configurations, scores them, and picks the best is performing a maximisation. The winner’s score is upward-biased by exactly the amount a maximum over K draws exceeds the mean, and no amount of care in the final measurement removes it if the final measurement uses material the selection has seen.

The apparatus. Before any candidate existed, deal banks were reserved by seed and their digests committed: a rolling hash over the six dealt hands and the dealer of every deal in the bank, computed without constructing a policy or playing a game. An adversary population was split in half by a rule fixed in advance, and one half was *sealed* — encrypted, with the plaintext’s SHA-256 committed — so that it could not be trained against even accidentally. The engine enforces the seal: a run that names a sealed seed fails unless an explicit unseal variable is set, which is set once, at the top of the final evaluation, and recorded.

The protocol. The entire final battery — every cell, its sample size, its threshold, the arithmetic behind the sample size, and the conditions that would count as failure — was written down and committed before any holdout bank had been played and before the sealed half had been decoded. The evaluation phase then ran from a cleared context, reading that protocol and nothing else from the project’s document set. Where it needed a training figure for comparison, it took it from the protocol, which states its expectations in advance for exactly that purpose, so that agreement is a check and disagreement is a finding.

Deviations. Anything the evaluation did that the protocol does not specify is recorded as a deviation, next to the affected result rather than in a footnote. There are 26 (Appendix E). A battery with recorded deviations and honest numbers is worth more than one that silently matched its own protocol.

7.6 The commit gate runs before any strength number

The failure this prevents is dissociation between win rate and soundness. This corpus contains a configuration that scores 56.60% against a prior version while carrying 44.83% provably-dead asks — asks for cards its own team already demonstrably owns — a dead run of hundreds of consecutive such asks, and 14% of its games killed by the action cap. It is stronger and it is broken.

So a configuration is gated *before* any strength number is computed for it, on eight rules covering dead asks, dead runs, action-limit games, the mirror game-length tail, late declarations, and the side-channel tests below. A configuration that fails is rejected regardless of its win rate. The gate is run on *training* material by design: it is a soundness check on self-play behaviour, not a strength claim.

A gate needs a negative control. The corpus maintains a configuration built to be rejected, and the gate must reject it; a gate that cannot would certify nothing about the one it accepts. §12.2 reports both.

7.7 Detection floors

An exploitability number is produced by a search, and a search that finds nothing has two possible explanations: there is nothing, or the search is too weak. A **detection floor** separates them. Plant an artificial handicap of known size in a target, fit a responder against the handicapped target, and measure the *excess* of that responder’s edge over its edge against the unhandicapped one. The smallest planted edge the procedure recovers is the floor.

Two findings from doing this are methodologically important and neither was expected.

1. **The floor does not buy down with evaluation games.** An early prediction that it would fall as $(\text{evaluation games})^{-1/2}$ was refuted at its own predicted value: at four times the games the floor is 1.53, and the sub-floor rung is still undetected — not because the interval is wide but because the excess resolves *to* zero. Below roughly a point and a half the responder is *empty*, not unresolved. The binding constraint is search power, not evaluation power.
2. **Floors are family-specific.** The declaration channel has its own, higher floor (2.13), and reading a declaration-objective result against the general floor would overstate it.

A class of adversary may contribute to an exploitability claim only after it has passed a planted-edge calibration at or below the effect size being claimed. That precondition is why the negative controls of §13.7 are run at all: an instrument that cannot recover a planted edge of a given size cannot report the absence of a real one of that size.

7.8 The homogeneity and side-channel gates

The problem. The team under evaluation is three identical agents with no shared secret. “Identical and deterministic” means their coordination is common knowledge (§3.2) — which is legitimate — but it also means an implementation detail can become an illegal signalling channel without anyone intending it. Before this cycle the constraint had only ever been checked by reading the source.

The apparatus. Four mechanical tests, run over all four engine decision types:

S3, listening substitution. Does a seat’s behaviour change more when a *teammate*’s hidden action changes than when an *opponent*’s does, beyond what the rules explain?

S4, stream independence. Is the seat’s private random stream drawn independently of the deal?

S5, posterior invariance. Does the seat’s action depend on the true card location beyond what its own posterior supports?

S6, seat isolation. Replay every decision from (own hand, public event stream, rules, reset seed) alone and count the decisions the reconstruction cannot reproduce. Zero tolerance.

The calibration, which is the part that makes it a test. Three positive controls were built, each a policy carrying one planted cheat, and each must fail exactly the tests the protocol names for it in advance and no others. §12.1 reports the result on holdout material. Without those controls, a clean report from the gate would be uninformative.

One honest limitation, stated wherever the gate is used. S6’s zero-tolerance form requires one thread — the audit depends on the deal-to-thread partition, and work-stealing does not fix that partition at any higher count — and agents rebuilt per deal. Rebuilding agents per deal *changes play*. So the gate certifies “with cross-deal residue removed, nothing else depends on anything it should not”. The residue is not removed from the shipped mode, and every strength number in this paper is measured under it (§13.8).

7.9 Worst case and minimax regret over a shared panel

The project’s standing reporting rule is that **the worst case leads and an aggregate never does**. A mean over an opponent panel rewards beating weak opponents by more, which is not the quantity anyone cares about.

Two statistics are reported, both over a panel scored identically for every arm:

- the **worst cell** per arm — the panel member on which it does least well;
- the **minimax regret** per arm — for each member, how far the arm falls short of the best arm on that member, maximised over members.

They can disagree, and in §13.1 they do. Reporting both, and reporting the panel mean last and labelled as a diagnostic, is the discipline; choosing whichever one flatters the release is the failure it prevents. Every arm must carry a cell for every member, or minimax regret is not defined at all — which is why the panel is fixed by name in advance and scored for all arms on the same banks.

A caveat the panel design forces. An arm that is itself a panel member meets itself and carries a free 0.00 cell, which can only flatter its worst case. Where that matters, §13.1 prints the worst case both with and without self-cells.

8 What “exploitable” was made to mean

The project’s standing reporting rule is that the worst case over the opponent panel leads and an aggregate never does. That rule is correct and it had run out of discriminating power for the comparison v0.7 has to make: a panel of hand-built archetypes measures robustness to the behaviours someone thought to build, and every arm in this corpus beats most of them by 24.72 to 50.00 points. A definition was therefore written before any v0.7 measurement was taken, and this section states the parts of it that constrain how the numbers in §13 and §14 may be read.

The unit is the team, not the seat. What is evaluated is the joint policy $\pi = (\pi_1, \pi_2, \pi_3)$ of three identical agents. The headline adversary controls *all three* opposing seats. A one-seat deviation column is reported alongside it on the same deals, because a three-seat upgrade and a one-seat upgrade are different quantities and the corpus has confused them before (§13.4).

The adversary is white-box, with an explicit exclusion list. It is given the target’s source, its frozen parameter vector, and unlimited offline access to it. It is *not* given the deal, the deal seed, any seat’s hand, or the engine’s confidence field. The exclusions are the point: a homogeneous team of three identical published deterministic agents inverts the usual assumption of the team-correlation literature, where the correlation device is the team’s own and unknown to the opponent. Here the device is common knowledge, and the adversary class built to exploit that — transcript inversion against a deterministic team — is the one §14 reports as *Zeight*.

Homogeneity is two prohibitions, and it is checked mechanically. The team gets no shared secret pre-play seed. Everything else a seat-conditioned policy can do is legal. An *illegal* side channel is defined by the bridge encrypted-signal criterion, made mechanical by a closed list of implementation-only discriminators plus four tests over all four engine decision types. Until phase 3 none of those tests existed as code; the constraint had only ever been checked by inspection. §7.8 reports the harness that implements them and the three planted cheats that calibrate it.

Every exploitability number is a lower bound. Reported as $\geq x$ with class, correlation regime, seat count and budget attached. Computing the true value is NP-hard even white-box. The consequence for this paper is stated once and then honoured everywhere: **a search that fails to find an exploit is evidence about the search, not a property of the target.** The v0.4 record contains exactly this reading error, turning a null result into a win, and §16 carries the correction forward rather than repeating it.

A class may contribute only after calibration. An adversary class is admitted to an exploitability claim only once it has passed a planted-edge calibration at or below the effect size being claimed. That precondition is why §7.7 exists and why the negative controls of §13.7 are run at all: an instrument that cannot recover a planted edge of a given size cannot report the absence of a real one of that size.

What the definition does not license. It does not license the claim that a policy passing every test is unexploitable, and it does not license reading the panel’s worst cell as an exploitability figure. The panel measures robustness across playstyles; the adversary search measures what a fitted responder can do. They are different objects and this paper keeps them in different sections.

9 Characterising the instrument, and searching for what beats the frontier

The first two phases of the v0.7 programme did no policy work at all. One rebuilt the measuring instrument and established what it can and cannot see; the other tried, in the open, to find anything that beats the v0.6 *frontier*. Both were run before any candidate architecture existed, and both returned results that reshaped what the rest of the programme could claim.

9.1 The frontier, and why the bar is not the deployed policy

v0.6 shipped a policy with its determinized search switched off, and also *measured* that search working. The set of configurations available at the end of that cycle is therefore not a point but a **frontier**: the deployed blueprint at one end, and progressively more expensive searching configurations at the other. Three points of it are named throughout this paper — the deployed policy **INCUMBENT**, the cheapest searching configuration on the frontier **F-cheap**, and a more expensive one **F-mid**.

This distinction is the single most consequential methodological choice in the v0.7 programme. Beating the deployed policy is the easy claim, and several unfitted one-switch deviations of it already do. **The bar was therefore fixed as F-cheap**, and §12.3 records that the protocol committed to it in advance.

9.2 What the instrument can see

The detection-floor apparatus of §7.7 was built and calibrated here, and produced the two findings that section reports: the floor is 1.53 points, and it does *not* buy down with evaluation games. Below roughly a point and a half a fitted responder is empty rather than unresolved, so the binding constraint on every exploitability statement in this paper is search power.

Two further instrument results matter for how later sections read.

Transcript inversion works, and it is not an attack. Because the target team is deterministic and published, an adversary can invert its transcript: for every ask the target made, ask what deals would have produced that ask, and contract the posterior accordingly. Measured, this contracts the deal posterior by ~ 2.0 bits per observed ask beyond the certificates the rules already supply — the first such measurement in this corpus, and one the threat model’s literature review found no external precedent for. Converting those bits into play is worth +1.52 points unfitted, the strongest single arm anyone built against the deployed policy.

It is not an exploit. The excess tracks planted-edge size within ± 0.4 on all eight rungs of the calibration ladder, which is the signature of a general strength gain rather than an exploitation instrument. And no linear handicap adds more than 0.07 bits per ask, so within this policy family there is *no readability handicap for a stochastic policy to buy*. A brief that asked for a stochastic policy conditional on the inverter biting was therefore closed on its antecedent rather than skipped.

A cost figure the corpus had wrong by two orders of magnitude. The record priced the searching configuration at $242\times$ the blueprint. That figure describes the *unrestricted* search, which is not the point of the frontier anything was measured at. The endgame-restricted operating point was re-measured at $\sim 3.2\times$. §13.9 corrects that correction in turn.

9.3 Open-ended adversary generation

The second phase’s constraint was that the searches must not share a bias — *fifteen runs of one search are one run*. Every exploiter search in the corpus before it had been the same search: a cross-entropy fit of a linear vector, maximising win rate, from one seed, against one target. The phase built six adversary classes, three correlation regimes, and a taxonomy of channels, and measured channel ceilings *before* fitting anything against them.

The headline is negative and it is the reason this paper exists in the shape it does. Nothing built exploits the v0.6 frontier beyond the detection floor. The best measured exploitability of the deployed policy remains an in-class figure of $+0.79$ [$+0.48, +1.10$] over 96,000 games — *below* the class floor of 1.53. Seven independently fitted searches confirmed it; none cleared its class floor.

Every arm that beat the deployed policy by more than the floor turned out, on measurement, to be a **better policy rather than an exploit**: the target’s own test-time search at $+1.89$, and a single out-of-class coordinate at $+2.71$ whose gain survives against opponents sharing none of the target’s machinery. The second of those is worth dwelling on, because it is where this cycle’s entire strength gain comes from: it is a *hand-set* coordinate found by an unfitted sweep, with no fit at all, and against the more expensive frontier point its edge is -0.99 — negative. It is a strength gain with a worst case, not an exploit.

A defect found in the target rather than an attack on it. The declaration-urgency channel the phase expected to headline has a measured ceiling of $+0.38$ [$-0.06, +0.81$] — dead as an attack. But the *same* channel is worth $+1.23$ to $+1.62$ points to a configuration that simply switches it off. The adversary owns the smaller half of a two-point defect in the target.

A cliff nobody can reach, and a termination duty inherited by anyone who removes it. Behind the incumbent’s event counter sits a 15.18-point cliff: past a threshold its declaration rule drops its team-ownership floor and cashes half-suits at close to a coin flip. No adversary reaches it — the longest game any of them produced against an unmodified incumbent is 149 events. But that escalation is also the policy’s termination guarantee, and a configuration that switches it off has a self-play tail of 405 events. Removing a defect here creates an obligation, and §11.1 is how it was discharged.

9.4 What these two phases established before any candidate was built

1. The v0.7 case rests on **beating the frontier, not on closing exploits**. There are no exploits above the floor to close.

2. The bar is **F-cheap**, fixed in advance.
3. The only thing found that beats the frontier is a configuration built from the frontier’s own search plus one hand-set coordinate plus a switch — a *configuration change*, not an architecture. This is the object called the phase-2 composite throughout this paper, and §15.1 is the section in which it becomes decisive.
4. The instrument’s floor is 1.53 points and will not improve by spending games.
5. The evaluation material is sealed, and the adversary population is split in half by a rule fixed in advance, with half of it encrypted (§7.5).

10 The hypotheses that were explored and rejected

Five candidate architectures were built in isolated worktrees, gated before they were scored, and certified against the side-channel definition by the harness of §7.8 rather than by inspection. **One survived, and it is the configuration change of §11 rather than an architecture.** The four that died are set out here with the measurements that killed them, because each closes a standing entry in the project’s design register with a number instead of an argument, and because a closed direction is the part of this cycle that does not depreciate.

Every figure in this section is on *training* material, at the budget stated with it, and none of it is certified: every pooled interval reported below has a lower bound under the 1.53-point floor of §7.7. They are measured edges, and the distinction is kept.

10.1 The declaration-allocation channel closes at exactly zero

The register’s second-ranked entry, priority 0.48, held that the whole of v0.6’s margin lives in the declaration channel and that roughly two points of it remained. Its named mechanism was that the deployed allocator — a product of marginals — picks a different allocation from the joint posterior’s argmax often enough to matter.

It does not. Over ~210,000 recorded voluntary declarations, across two banks and both urgency arms, the joint argmax and the marginal-product argmax select *the same allocation at every one*, with the joint score holding a strict opinion on 95.6%–100% of genuinely ambiguous cases. The reason is algebraic rather than statistical: the sequential joint’s first chain-rule factor *is* the marginal, and the re-normalisation that follows reweights the survivors without reordering them.

Two further measurements close the direction rather than merely the mechanism. First, the *exact* joint maximiser is a worse decision rule than the deployed approximation on the declaration decision itself: paired, it fixes 218 declarations and breaks 454, a net -1.313 pp, which converts to about -1.58 win-rate points. This is the second independent instance in this corpus of the project’s standing habit paying off — “exact Bayesian inference” is an assumption to test, not a property to assume — the first being v0.6’s finding that the exact posterior is a worse predictor of card location than the approximation the policy deploys. Second, 72.4%–74.4% of the error class sits in flat-posterior states where the exact object is provably a coin flip, so most of what looked like headroom is an information limit and not a policy defect.

And the channel is largely downstream of a defect that the configuration change removes anyway. Deleting the urgency escalation removes 71% of the entire allocation-error class and drops the oracle ceiling for a *perfect* allocator from 2.45 win-rate points to 0.84 — below the detection floor. The consequence is a composition constraint that this paper honours: a declaration-channel gain and the urgency-off gain may never be added, because they are the same defect counted twice.

10.2 A leaf evaluator fitted 74× better buys nothing

The inherited conditional held that the test-time search cannot be judged until its leaf evaluator is rebuilt, because the shipped one is algebraically close to a rescaling of the hit probability. A leaf was fitted for the purpose, and the statistic that matters — *between-candidate* R^2 , the only component a paired deviation rule can consume — improved from 0.00108 to 0.07976, a 74× gain replicated out of sample on $\sim 1.3\text{M}$ leaves.

In play it is worth +1.84 against the material control’s +2.01, losing on both banks individually. The diagnosis is conversion failure, not fit failure: the rule’s deviation rate is 0.3205 for the material leaf and 0.3237 for the retuned one, so the deviation rule at the shipped shrinkage is **nearly leaf-invariant**. A better leaf is not the lever.

The conditional itself is discharged separately, and by re-measurement rather than by the new evaluator. The single underpowered cell the conditional rested on — one 4,000-game cell at +0.08 — re-runs at +2.19 replicated in the endgame regime with v0.6’s *own* leaf. Full-game truncated search already works with the leaf the conclusion said could not support it; it is simply at a dominated operating point.

10.3 Fitting against a per-decision objective produces a worse policy

The register’s enabling entry held that a per-decision objective is necessary because the scoreboard is too noisy to resolve small effects. **Its precision claim is confirmed and its fitting form is killed**, and the two halves are independent.

The design effect of deal clustering was measured for the first time in this cycle, and a one-point-equivalent effect resolves in 284–560 games on the declaration channel against 9,604 on the scoreboard: the precision claim confirmed at 17×–39×.

But all three self-oriented per-decision objectives, fitted at matched budget against a per-game control, produced *worse* policies: -2.26, -1.59 and -1.20 points, every one replicated in sign and two of them clearing the floor as certified negatives. Each fit moved its own proxy in the intended direction and lost anyway: the declaration-oriented fit bought +0.39 pp of declaration accuracy by paying -2.33 pp of ask accuracy, and the ask-oriented fit did the reverse. The two per-decision channels trade against each other, so a fitter holding 55 coordinates buys one out of the other. Keep the estimator; drop the objective.

10.4 Nothing learned survives a free random tie-break

53.80% of contested ask decisions in v0.6 tie bit-for-bit. The open question was whether anything can be *selected* inside that group, as opposed to merely randomised. A learned re-ranker was built — the corpus’s first learned agent — restricted to the tie group.

Against the deployed policy it scores +1.19 pooled, which is indistinguishable from the free hash tie-break’s +1.14; head to head against that free tie-break at 48,000 games it is -0.01 ± 0.45 , replicated. **Nothing learned survives the free random tie-break**. The correct control for anything operating inside the tie group is the decorrelated policy and not the deployed one, which is a methodological point this paper inherits and applies.

The same capture explains why. Setting every candidate’s true advantage to exactly zero and feeding the capture’s real standard errors through the engine’s real deviation rule reproduces the search’s observed deviation rate to a 0.4% relative error on a quantity the null was never fitted to: the search’s per-decision selection signal is $\sim 84\%$ winner’s curse on its own Monte-Carlo draw, and $\sim 95.5\%$ inside the tie group, where the rule applies no shrinkage at all.

10.5 Two directions closed on arithmetic before they were built

Forced-endgame allocation. The register priced closing the entire forced-endgame accuracy gap. Re-derived from this cycle's own capture it is worth 0.030 win-rate points — the register's own figure corrected *upward* by a factor of two and still a fiftieth of the detection floor. The one route by which it could have returned, an adversary driving incidence up, was measured and costs the adversary 17 points.

A stochastic policy bought to reduce readability. The brief made this conditional on the white-box inverter biting. Transcript inversion contracts the deal posterior by ~ 2.0 bits per observed ask — the first such measurement in this corpus, and one the threat model's literature review found no external precedent for — and converting those bits into play is worth +1.52 points unfitted. But the excess tracks planted-edge size within ± 0.4 on all eight rungs, which is the signature of a general strength gain and not of an exploitation instrument. And no linear handicap adds more than 0.07 bits per ask, so within this policy family **there is no readability handicap for a stochastic policy to buy**. The antecedent fails and the limb is closed rather than skipped.

10.6 What the four kills leave

The register the cycle inherited described a policy with several points of accessible headroom in four named places. Three of them are now closed with measurements and the fourth is confirmed as an instrument rather than a lever. What remains as the located gain is a single coordinate found by an unfitted sweep — not by any of the architecture work — and §13.3 measures it on holdout. That is the barrier this paper names, and §19 states it as such.

11 What v0.7 changes

v0.7 is **a configuration of the v0.6 policy family, not a new architecture**. That sentence is the protocol's, written before the evaluation ran, and the evaluation did not disturb it. The frozen object is v0.6's fitted 37-coordinate vector, one hand-set extended coordinate, three switches that turn off machinery an earlier phase measured as defective, one new mechanism, and the truncated test-time search that v0.6 shipped switched off:

```
v07:r12=25,rtie=1,pool=-1,oppfloor=-1,force=1000000,askfloor=-1,stall=12,s1=1,
det=12,cand=4,kappa=2.5,rbelief=indep,depth=12,maxq=26
```

It is written to `engine/fishbot\_v07.json` by `engine/freeze\_config\_v07.py`, which *executes* its round-trip assertion rather than printing it: the evaluation reconstructed the spec from that file and never typed it. The artifact carries 55 resolved coordinates and the mirror pathology digest 5f81f440fc9c272a87e87c05fecc7b74.

11.1 The five keys, and what each was expected to be worth

r12=25 — half-suit contestation. The information-denial coordinate `oppCertDonate`, found by an *unfitted* coordinate sweep in phase 2 rather than by any architecture work. Phase 2 measured a single hand-set setting of it at +2.71 points against the deployed policy over 48,000 games with no fit at all. The protocol states in advance that this key carries the gain and that a holdout leave-one-out drop of roughly two points is the check: +2.10 on training material. §13.3 reports what holdout gives.

The endgame-truncated search. `s1=1,det=12,cand=4,kappa=2.5,rbelief=indep,depth=12,maxq=26` — machinery v0.6 built, measured and shipped *off*. It is a determinized search that reconstructs all six continuation players at their own information sets rather than making them clairvoyant, with a paired lower-confidence-bound deviation rule. Its leave-one-out value on training material is +0.77. **rtie=1** — **decorrelating the tie group.** 53.80% of contested ask decisions in v0.6 tie bit-for-bit and are decided by an unstable `std::sort` order. This key replaces that order with a hash of the public event stream. Training leave-one-out +0.20.

pool=-1,oppfloor=-1,force=1000000,askfloor=-1 — **the urgency escalation off.** These switch off v0.5's 220-event declaration clock. Behind that clock sits a 15.18-point cliff: past the threshold the declaration rule drops its team-ownership floor and cashes half-suits at close to a coin flip. No adversary ever reached it — the longest game any phase-2 adversary produced against an unmodified incumbent is 149 events — but a candidate that simply switches the clock off inherits its termination duty. Training leave-one-out +0.02.

stall=12 — **the replacement termination guarantee.** Escalate after twelve consecutive public events in which this seat's hard-certificate hash does not change. The threshold is established at zero game cost: over 800 mirror games v0.6 never produces a no-progress run longer than 6, and a *deadlocked* configuration — the gate's negative control, not FROZEN — produces 326. Any threshold in [12, 60] is simultaneously unreachable in ordinary play and immediate in a deadlock; the separation between the two regimes is fifty-fold, though the threshold's own margin over observed ordinary play is twelve against six. Armed, its mirror digest is byte-identical to the same configuration without it: the rule's cost in ordinary play is not small, it is identically zero, because it never runs.

The last three are individually indistinguishable from zero at the margin. *As a group* they were measured at +0.78 [+0.33, +1.22] on training material, replicated on both training banks, as a paired head-to-head against the phase-2 composite. The protocol forbids attributing that group's value to any one of them, and §13.3 keeps that discipline. One further key, `m2=0`, was dropped from the freeze because its leave-one-out drop is exactly zero; it is measured anyway, because the ablation lattice preregisters the cell.

11.2 What was deliberately not changed

No fit of any kind was applied to the frozen configuration. Its vector is v0.6's, found by v0.6's repaired optimiser under a minimax-regret objective over a four-member panel. The architecture work of phase 3 produced five candidates and contributed one of them — the defect stack above — and contributed it as a configuration change plus one small mechanism rather than as an architecture. §10 is what happened to the other four, and it is the more valuable half of the cycle.

The engine defect that the evaluation runs *under* is also deliberate. Agents are constructed once per thread and reused across every deal that thread is handed, and some per-agent state survives `reset()` and is reachable only under the search. Phase 4 named the carrier — `V06Agent::roll`, the rollout engine, which `reset()` never touches — and did not fix it, because the one-line repair changes the play of every searching configuration by about one deal in 800, and shipping that after the freeze is the one thing a freeze exists to prevent. It is a named, quantified engine defect, every strength number in this paper is measured under it, and §13.8 measures it again on holdout material.

Table 1: The deal banks, each digest recomputed before any cell was played by folding the six dealt hands and the dealer of every deal into a rolling hash without constructing a policy or playing a game. The committed digests were computed by a *different build* of the engine at seal time, so agreement is between two binaries as well as two dates.

Bank	Deals	Digest reproduced	Matches commitment
7090001	24,000	896dbc89be124d85	✓
7090002	24,000	0b6e40d834ac0ca1	✓
7090003	24,000	863bea69baf6e73c	✓
7090004	24,000	54f257c3f8ae9fab	✓
7090005	24,000	268a1dae71a31713	✓
7091001	24,000	958ada042cc26900	✓
7091002	24,000	5c39af3b5e0bd9a0	✓

Table 2: The calibrated side-channel positive controls, on holdout material. *Must fail* is the set the protocol names for each planted channel in advance — not one test each; the seed cheat is visible to two of them. A cheat failing a test outside its own set, or missing one inside it, would mean the gate does not discriminate and no seat-isolation statement in this paper could be read as a measurement.

Planted channel	S3	S4	S5	S6	Must fail	As required
v07x:cheat=seed	PASS	FAIL	FAIL	PASS	S4, S5	✓
v07x:cheat=shared	PASS	PASS	PASS	FAIL	S6	✓
v07x:cheat=conv	FAIL	PASS	PASS	PASS	S3	✓

12 The protocol, and the seal

The evaluation was run from a cleared context against a protocol committed before any holdout bank had been played and before the sealed adversary half had been decoded. The phase read that protocol and nothing else from the v0.7 document set: not the candidate report, not the adversary report, not any training log. Where a training measurement is quoted for comparison it is quoted *from the protocol itself*, which states its expectations in advance for exactly that purpose — so agreement is a check and disagreement is a finding.

12.1 The material, and what had to be verified before anything was measured

All 7 bank digests match (all), at 24,000 deals each. The sealed adversary half decodes to 14 rows whose plaintext SHA-256 matches the commitment recorded at **f4581da**. The freeze artifact verifies (PASS), its spec rebuilds character-identically from its own ordered option map, and its round-trip *with the search on* is identical over 800 games. The engine’s own audit reports 0 soundness violations in 6,737,436 checks and its self-test passes on 424,986 checks.

The homogeneity gate of §7.8 is calibrated here: each planted channel fails exactly the tests the protocol names for it and no others, identically on both banks, over 12 cells (Table 2).

Three of the fourteen sealed adversaries are policies the rest of the panel already carries. They are run anyway, because the protocol fixes the panel by name, and their agreement with the corresponding scripted or frontier cell is used as a free determinism check (§16.2).

Table 3: The commit gate, eight rules, run on training material by design — it is a soundness check on self-play behaviour, not a strength claim. The negative control is the configuration the corpus built to be rejected; a gate that could not reject it would certify nothing about the one it accepts.

Rule	Frozen	Incumbent	F-cheap	Negative control
provably-dead asks $\leq 0.10\%$	ok	ok	ok	fail
longest dead run ≤ 5	ok	ok	ok	fail
games with a run $\geq 6 = 0$	ok	ok	ok	fail
action-limit games $= 0$	ok	ok	ok	fail
mirror tail max < 220 , p99 ≤ 150	ok	ok	ok	fail
declarations at/after event 220 $= 0$	ok	ok	ok	ok
S3/S4/S5 certified, zero tolerance	ok	ok	ok	ok
S6 $= 0$ at <code>-threads=1 -freshagents</code>	ok	ok	ok	ok
verdict	PASS	PASS	PASS	FAIL

12.2 The commit gate runs before any strength number

A configuration that scores higher while failing the pathology rules is rejected, and the ordering is not a formality: the corpus contains a configuration that scores 56.60% against v0.4 while carrying 44.83% dead asks and losing 14% of its games to the action limit. Win rate and soundness are dissociable in this family.

FROZEN passes all 8 rules: 0.05820% provably-dead asks against a 0.10% bar, longest dead run 1, mirror tail max 134 and p99 130, and 0 seat-isolation mismatches in 542,483 audited decisions. Reported and not gated: 99.805 events per game, 50.716% ask hit rate, 2.556% misdeclaration. The negative control fails 5 rules — G1, G2, G3, G4, G5 — which are exactly the ones the protocol names in advance, on 2.59217% dead asks, a longest dead run of 326 and a mirror tail reaching 405.

12.3 What the protocol said would count as failure

Seven conditions were written down in advance, “because after the numbers are in every one of these has a reading that makes it sound like something else”. They are reproduced in full in Appendix C and answered condition by condition in §15. Two of them came out against the release.

The protocol also fixed the primary bar. F-cheap is named rather than the deployed policy, deliberately: the deployed policy ships its search off, beating it is the easier claim, and an earlier phase had already shown that several unfitted one-switch deviations of the incumbent do it. The v0.7 case had to be against the frontier.

And it fixed, in advance, the sentence to be used if the partner-regime ordering survived — naming the direction so it could not be chosen afterwards, and recording that on training material that comparison goes *against* the release. §13.4 reports what happened and uses the sentence.

12.4 Deviations are recorded next to the result they affect

The protocol requires a deviation for anything the evaluation did that it does not specify: a cell dropped, a cell added, a threshold read a particular way, a command whose flags differ. There are 26 of them, 3 adding cells or checks the protocol’s own questions turn out not to be computable without, and 3 of them corrections. Appendix E is the full register, read out of the results document rather than retyped. 0 preregistered cells were dropped and 18 were added — the added ones because two of the protocol’s own questions are not computable from the cells it names (Table 22).

Table 4: Worst cell per arm over the shared panel. This is the headline and it is printed first.

Arm	Worst cell	Edge (pts)	95% CI	Per bank
FROZEN	composite	-0.04	[-1.41, +1.33]	7090001:+1.67 7090002:-1.75
INCUMBENT	SEALED:X01xC3f	-4.24	[-4.86, -3.62]	7090001:-3.86 7090002:-4.63
F-cheap	composite	-2.08	[-3.48, -0.69]	7090001:-2.21 7090002:-1.96
composite	composite	+0.00	[+0.00, +0.00]	7090001:+0.00 7090002:+0.00

13 Results

Every measurement in this section comes from the artifacts listed in Appendix F and is emitted into this document by `engine/build\_tables\_v07.py--paper`. The one exception is labelled where it occurs: the two *training* rows of Table 11 are quoted from the protocol, which states them in advance, and are the only hand-entered numbers in the section. The battery is 428 scored cells over 4,322,400 games with 0 audit violations and 1 cell carrying any action-limit hit.

The order of this section is fixed by the project’s reporting rule and not by what flatters the release: **the worst case and the minimax regret come first, the aggregate comes last and is labelled a diagnostic, and the partner-regime table appears whether or not it helps.**

13.1 The panel: worst case and minimax regret

248 cells. Four arms — FROZEN, INCUMBENT, F-cheap and the phase-2 composite — each scored against the *same* 31-member panel on the *same* two banks under the same protocol: 19 near-class members at 6,000 deals a bank, 10 far-class archetypes at 1,500, and 2 expensive members at 1,200. All four arms carry a cell for all 31 members, which is what makes minimax regret defined at all.

FROZEN’s worst cell is -0.04 [-1.41, +1.33], against INCUMBENT’s -4.24 [-4.86, -3.62]. The preregistered secondary claim that the worst case is not catastrophic passes.

That cell does not replicate, and this paper’s own rule says to say so. FROZEN’s worst cell is +1.62 on 7090001 and -1.75 on 7090002, and it is attained at a different member on each (SEALED:X01xC3f and composite). On one bank FROZEN has no negative cell anywhere on the panel. Under §7.2’s sign rule the pooled -0.04 is *not replicated* (NO), and the same criticism this paper makes of INCUMBENT’s regret-attaining member below applies to its own headline cell. What survives replication is the comparison the protocol actually gated on — FROZEN’s worst cell is far above INCUMBENT’s on both banks — not the value -0.04 itself.

One caveat belongs beside that comparison rather than beneath it. INCUMBENT, F-cheap and the phase-2 composite each meet *themselves* in the panel and so carry a free 0.00 cell, which can only lower a worst case; FROZEN is not a panel member and carries none. Excluding self-cells, the phase-2 composite’s worst becomes +1.23 against SEALED:X01xC3f and the other three are unchanged, so the S6 claim survives on that reading too. **But the phase-2 composite has the better worst cell on both readings** — +0.00 with its free self-cell and +1.23 without — so FROZEN does not have the best worst case of the four arms, only a far better one than the arm the protocol asked it to be compared against. Both readings are printed because only printing the flattering one would be a choice.

FROZEN is **3rd of four on minimax regret**, at 4.53 against F-cheap’s 4.00 and the phase-2 composite’s 4.08. The arm with the best worst case is not the arm with the best regret, and this paper does not resolve that tension by choosing the metric that wins.

The regret is concentrated in the far archetypes, where the blueprint arms beat FROZEN outright

Table 5: Minimax regret over the shared panel, best first. The protocol sets no threshold on this quantity; it requires it to be reported and to lead. It does not favour the frozen configuration.

Arm	Minimax regret (pts)	Attained against
F-cheap	4.00	v03
composite	4.08	withholder
FROZEN	4.53	withholder
INCUMBENT	5.67	SEALED:X01xC3f

Table 6: The far archetypes, where the regret difference lives. Every arm beats every one of these by between 24.72 and 50.00 points; the differences between arms are what minimax regret is measuring.

Far archetype	Frozen	Incumbent	F-cheap	Composite
v03	+28.72	+25.47	+24.72	+28.40
v02	+34.32	+31.68	+32.67	+33.17
lockout	+30.78	+28.65	+28.97	+29.07
detective	+29.65	+28.08	+27.98	+28.32
hunter	+46.93	+48.58	+48.65	+47.10
diversifier	+43.40	+44.90	+45.18	+43.32
bluffer	+49.87	+49.98	+49.95	+49.88
silent	+32.75	+34.23	+34.05	+30.92
withholder	+25.35	+29.88	+29.55	+25.80
random	+50.00	+50.00	+50.00	+50.00

— on opponents every arm already beats by 24.72 to 50.00 points (Table 6). Against the hardest sealed adversary the ordering reverses: on SEALED:X01xC3f, FROZEN is +1.43 [+0.80, +2.06] where INCUMBENT is -4.24 and F-cheap is -1.25. 2 of the four arms beat that member with a lower bound above zero (FROZEN, composite); the phase-2 composite takes +1.23 [+0.60, +1.86].

INCUMBENT’s attaining member is decided by 0.004 points over the runner-up, against a half-width orders of magnitude larger, and it flips between banks: the regret *value* is stable, the name attached to it is not.

The aggregate, printed last and as a diagnostic. Over the 31 shared cells the panel means are FROZEN +14.94, the phase-2 composite +14.35, F-cheap +13.28 and INCUMBENT +11.99. The rule forbids leading with this and it has not been led with. It points the opposite way from minimax regret, and both are in this paper because neither on its own is what the panel measured. The full 31×4 table is Appendix D, so a reader can take the worst cell themselves rather than take this paper’s word for which it is.

13.2 Strength against the frontier

The primary claim: FROZEN against F-cheap is +3.33 points [+2.88, +3.78] over 48,000 games, sign replicating on both banks (+3.67 and +2.99), lower bound above the pre-committed 1.53 reference bar, gate clean and every verification precondition passed. **Verdict: CERTIFIED advancement.**

It also beats the deployed policy by +4.63 [+4.19, +5.06], F-mid by +2.89 [+2.00, +3.78] — clearing the restriction the protocol placed on that cell in advance — and v0.5 by +5.18.

The composite row is a failure and §15 is where it is treated as one. FROZEN does not beat the phase-2 composite: +0.15 [−0.29, +0.59], an interval containing zero on both banks

Table 7: Headline strength against the v0.6 frontier, paired duplicate, both primary banks. The preregistered primary claim is the **F-cheap** row.

Cell	Opponent	Edge (pts)	95% CI	Games	Sign replicates
B2.1	v06	+4.63	[+4.19, +5.06]	48,000	yes
B2.2	F-cheap	+3.33	[+2.88, +3.78]	48,000	yes
B2.3	F-mid	+2.89	[+2.00, +3.78]	12,000	yes
B2.4	the phase-2 composite	+0.15	[−0.29, +0.59]	48,000	yes
B2.5	v05	+5.18	[+4.56, +5.81]	24,000	yes

Table 8: The attribution lattice. Add-one-in is measured from the deployed policy; leave-one-out is the drop from FROZEN. Reference cell FROZEN against INCUMBENT on the lattice banks: +4.73 [+4.10, +5.36].

Component	Add-one-in from v0.6	95% CI	Leave-one-out drop	95% CI
r12=25	+2.18	[+1.55, +2.81]	+1.69	[+0.80, +2.57]
the search	+1.91	[+1.38, +2.44]	+0.57	[−0.33, +1.46]
rtie=1	+0.71	[+0.08, +1.34]	+1.08	[+0.20, +1.97]
urgency-off	+1.37	[+1.12, +1.63]	+0.64	[−0.25, +1.53]
stall=12	+0.00	[+0.00, +0.00]	+0.00	[−0.89, +0.89]
m2=0 (<i>not in the freeze</i>)	+0.77	[+0.66, +0.88]	—	—

positive.

Declaration accuracy is reported and carries no claim. The declaration-family floor of §7.7 applies to the declaration-accuracy column here. The harness emits declaration accuracy as a bare rate with *no interval of any kind*, so no interval can be quoted for it and no claim can be made about the declaration channel. FROZEN runs 0.970–0.982 across the five cells against opponents’ 0.976–0.986. That is a diagnostic. Declining to claim is what discharges the rule; clearing a bar is not.

13.3 Attribution: is the gain located?

The protocol’s first advance prediction — that **r12=25** carries the gain, checkable as a leave-one-out drop of “roughly two points” — **holds on that check**: the holdout drop is +1.69 [+0.80, +2.57], with the training figure +2.10 inside the interval, and an add-one-in of +2.18. The stronger sentence — that this coordinate *is* where the gain lives — is not supported: the drop’s lower bound is +0.80, and §15.2 shows the protocol’s own one-third location test failing on the lattice’s primary bank.

The second holds exactly: **stall=12** is worth +0.00 [+0.00, +0.00] added and +0.00 removed — bit-identical, reproduced on holdout, in win rate and in events per game alike. The rung did not fire. The third is only half testable here: **m2=0** is worth +0.77 [+0.66, +0.88] added to the blueprint on holdout — an interval excluding zero, and larger than **rtie=1**’s add-one-in — while its *zero leave-one-out drop* was measured on training material and is not re-measured here, because the freeze does not carry the key and the protocol says in terms not to add it back. There is no **L-m2** cell and there could not be one.

The naive sum of the five components the freeze carries is +6.18; the measured whole is +4.73. The sub-additivity ratio is 0.766 [0.599, 0.933] — propagating the reference cell’s half-width and the five add-one-in half-widths in quadrature — and the phase-2 composition figure 0.83 lies inside it, so the two phases’ composition is not distinguishable here. **Quoting the sum would overstate**

Table 9: The partner-regime table against a v0.5 opponent. Team A is [arm, P, P]; team B is three copies of the opponent. Every delta carries an interval; the phase-5 results document compresses these out of its own table and they are restored here (§16.2).

Partners	Frozen – Incumbent	95% CI	Per bank		Sign replicates
itself	+4.49	[+3.61, +5.38]	7090001:+4.18	7090002:+4.81	yes
v06	+1.93	[+1.04, +2.82]	7090001:+2.07	7090002:+1.79	yes
v05	+0.78	[−0.09, +1.66]	7090001:+0.95	7090002:+0.62	yes
v04	+1.34	[+0.46, +2.22]	7090001:+1.30	7090002:+1.38	yes
v03	+1.40	[+0.56, +2.25]	7090001:+1.08	7090002:+1.72	yes
detective	+1.83	[+0.97, +2.68]	7090001:+1.35	7090002:+2.30	yes
withholder	−0.19	[−1.04, +0.67]	7090001:+0.00	7090002:−0.38	no
lockout	+1.89	[+1.04, +2.74]	7090001:+1.39	7090002:+2.39	yes

Table 10: The same table against a v0.6 opponent.

Partners	Frozen – Incumbent	95% CI	Per bank		Sign replicates
itself	+4.55	[+3.92, +5.18]	7090001:+4.88	7090002:+4.21	yes
v06	+2.75	[+2.14, +3.37]	7090001:+2.97	7090002:+2.54	yes
v03	+1.46	[+0.63, +2.29]	7090001:+1.25	7090002:+1.67	yes
detective	+2.34	[+1.49, +3.19]	7090001:+2.42	7090002:+2.26	yes

the configuration by 1.45 points. The preregistered attribution claim passes: the components are sub-additive and no single leave-one-out drop exceeds the whole.

The protocol’s own location test does not replicate, and §15.2 treats that as the failure the protocol says it is.

13.4 Partner regimes

This table appears whether or not it flatters the release, and it does not entirely.

6 of 7 changed-partner rows are positive and the minimum is −0.19, well above the −1.00 collapse threshold. The single negative row does not replicate across banks and is reported as not replicated. The preregistered claim that the advantage does not collapse under partner change **PASSES**. Against a v0.6 opponent, where the protocol predicts more stability, all four rows are positive and clear zero (Table 10).

The sentence the protocol pre-committed to, and it applies. The protocol states in advance that if the incumbent-baseline ordering survives the fuller measurement, the honest sentence is “*v0.7’s advantage transfers across partners about as well as v0.6’s did, and on the median-to-self-play ratio slightly less well*” — not “it transfers”. The ordering survived (Table 11): 0.313 against 0.380. So that sentence stands as this paper’s.

Beside it belongs the fact the ratio omits: v0.7’s absolute changed-partner median is +1.40 against v0.6’s +0.26, and the ratio is smaller mainly because the self-play row it divides by is more than six times larger. Both are true, and neither is the headline alone.

A one-seat upgrade among v0.6 partners is worth +1.93 against a v0.5 opponent (predicted +1.26) and +2.75 against a v0.6 opponent (predicted +2.88). That a single seat buys a large fraction of the three-seat gain is the strongest evidence in this paper that the gain is individual rather than conventional.

Table 11: Transfer across partners, v0.7 over v0.6 against v0.6 over v0.5. The two holdout rows are the same eight settings on the same two banks and are generated from the artifacts. The two training rows are quoted from the protocol, which states them in advance; they are on training banks, and the protocol records that its v0.6-over-v0.5 training row is computed on three of the seven changed settings only, so that row is not like-for-like with the holdout one above it. Reported and *not* gated, as the protocol requires.

Comparison	Min	Median	Self	Median/self
v0.7 over v0.6 — holdout	−0.19	+1.40	+4.49	0.313
v0.6 over v0.5 — holdout	−0.40	+0.26	+0.69	0.380
v0.7 over v0.6 — training, stated in advance	−0.15	+1.26	+2.94	0.428
v0.6 over v0.5 — training, stated in advance	−0.61	+0.64	+1.35	0.472

Table 12: Cross-play. Three runs of the frozen architecture, independently trained on disjoint fitting banks with different search trajectories and, for one, a different starting basin; pairwise parameter distance 6.3–9.6 over 55 coordinates.

Run \ partners	xp1	xp2	xp3
xp1	+4.54 [+3.91, +5.17]	+3.80 [+3.17, +4.42]	+4.55 [+3.92, +5.19]
xp2	+4.93 [+4.29, +5.56]	+4.91 [+4.29, +5.54]	+4.83 [+4.20, +5.46]
xp3	+5.15 [+4.52, +5.78]	+4.06 [+3.44, +4.69]	+4.20 [+3.56, +4.83]

13.5 Cross-play between independently-trained runs

Diagonal mean +4.55, off-diagonal mean +4.55, gap +0.01 against a per-cell half-width of 0.63. The worst off-diagonal cell (+3.80) is 0.40 points below the weakest diagonal cell (+4.20), and the best off-diagonal (+5.15) exceeds every diagonal cell. The preregistered claim that the advantage is not a private convention **PASSes** against a threshold of 1.50 points. For scale, the protocol cites the cooperative-play literature’s convention collapses — self-play scores in the twenties falling to low single digits under cross-play — as the phenomenon to look for. Nothing of that kind is present.

Head to head, so that “these are different policies” is measured rather than assumed: 2 of the three pairs separate from zero (Table 13).

13.6 Rule dialects

Every row keeps its sign, every row replicates, and the largest excursion from the default row is +0.78 on arb=turn — inside the 2.00-point tolerance. The preregistered claim that the edge survives the rule dialect **PASSes**. It survives 48-card Literature, all three arbitration orders, both declaration-permission changes and the full legacy dialect.

Two readings belong with the table. `-maxasks=360` is *bit-identical* to the default (yes): the 400-ask cap is never reached in this population, so the row measures nothing and the legacy residual below is arithmetically legacy minus *two* effective components rather than three. And that residual — legacy minus its isolable components, which the protocol assigns to the forced-endgame willingness ladder because it has no flag of its own — is −0.292 [−1.69, +1.11], an interval 9.6× the size of the residual. The data are equally consistent with the ladder contributing nothing.

13.7 The negative controls

If these misbehave, nothing else in this paper is a measurement.

Table 13: The three runs head to head, establishing that they are distinguishable policies.

Pair	Edge (pts)	95% CI
xp1 vs xp2	+0.20	[−0.43, +0.83]
xp1 vs xp3	+0.70	[+0.08, +1.32]
xp2 vs xp3	+1.05	[+0.43, +1.68]

Table 14: The rule-dialect table, unbundled. Both arms play the same dialect in every row. This is the first dialect sweep of its kind in the corpus and has no training-bank counterpart to be checked against.

Dialect	Edge (pts)	95% CI	Per bank	vs default
default	+4.73	[+4.10, +5.36]	7090003:+4.57 7090001:+4.88	+0.00
no-out-of-turn	+5.25	[+4.63, +5.87]	7090003:+4.72 7090001:+5.78	+0.52
no-cardless-declare	+4.90	[+4.27, +5.52]	7090003:+4.71 7090001:+5.08	+0.17
maxasks=360	+4.73	[+4.10, +5.36]	7090003:+4.57 7090001:+4.88	+0.00
arb=high	+4.85	[+4.21, +5.49]	7090003:+5.03 7090001:+4.67	+0.12
arb=turn	+5.51	[+4.88, +6.14]	7090003:+5.26 7090001:+5.77	+0.78
sets=8	+4.13	[+3.52, +4.73]	7090003:+3.77 7090001:+4.48	−0.60
legacy	+5.12	[+4.50, +5.75]	7090003:+4.62 7090001:+5.63	+0.40

Two of the three supra-floor rungs are recovered with a *lower bound* above the 1.53 floor. The third, the 0.08 rung, recovers +2.38 [+1.50, +3.26] — above the floor on the point estimate but with an interval that straddles it, so on the lower-bound reading this paper applies to its own primary claim it is *not* recovered (no). Its planted cost, +1.59 [+1.05, +2.13], straddles the floor too, so the rung is only supra-floor on a point estimate in the first place. The recovered size does track the planted size monotonically across all four rungs, and the ladder is sharply non-linear, with most of the damage arriving between the second and third. The sub-floor rung is **not** recovered — +0.70 [−0.20, +1.59], below the floor with an interval containing zero — which is the control that has to fail and does.

Two qualifications belong beside it. An earlier phase measured this same excess at −0.01 [−0.45, +0.42] and this evaluation gets +0.70: the same conclusion from a point estimate seven tenths of a point higher. And its upper bound *grazes* the floor. The verdict is unchanged — the test is whether the edge is recovered, and it is not — but the margin is thinner than the earlier phase’s, and a reader entitled to the caveat is entitled to it here rather than in a footnote.

The identity control is exact: FROZEN against itself is 0.500000 with a zero-width interval on both banks over 12,000 games each (yes). The fitted responders lose to *unhandicapped* FROZEN in all four rows, reproducing §14’s finding from a different fitting bank and against a handicapped target. The objective is *not* independent — these responders and four of the eight adversary searches both maximise win rate — so this is a second sample, not a second method. And the three planted side channels each fail exactly the tests the protocol names for them and no others (Table 2).

13.8 The seat-isolation residual on holdout material

The gate column is zero for every arm on both banks (0 nonzero cells). There is no new side-channel finding and the headline the protocol reserved for one does not apply.

The first column reproduces the known engine defect on holdout material and reproduces its shape: 4 cells are nonzero and every one of them belongs to a *searching* arm (yes), with the blueprint incumbent zero on both banks. FROZEN is zero on one bank and nonzero on the other, which the

Table 15: Planted-edge recovery. The recovered excess is the responder’s edge against the handicapped target minus its edge against the unhandicapped one. The sub-floor rung must *not* be recovered.

Planted hstr	Planted cost	Responder vs handicapped	Responder vs Frozen	Recovered excess
0.05 (<i>sub-floor</i>)	+0.78 [+0.25, +1.32]	−3.62 [−4.26, −2.99]	−4.32 [−4.94, −3.69]	+ 0.70 [−0.20, +1.59]
0.08	+1.59 [+1.05, +2.13]	−2.29 [−2.92, −1.67]	−4.67 [−5.30, −4.05]	+ 2.38 [+1.50, +3.26]
0.11	+8.52 [+7.93, +9.12]	+2.52 [+1.89, +3.16]	−6.90 [−7.52, −6.28]	+ 9.42 [+8.54, +10.31]
0.15	+11.44 [+10.85, +12.03]	+7.30 [+6.68, +7.93]	−5.34 [−5.97, −4.71]	+ 12.64 [+11.75, +13.53]

Table 16: The seat-isolation residual, 16 cells over 19,200 audited deals. The gate column is the preregistered zero-tolerance condition; the first column is the shipped mode.

Arm	Bank	-threads=1	-threads=1	-freshagents
FROZEN	7090001	0 / 813,512		0 / 813,512
FROZEN	7090002	3 / 816,049		0 / 816,016
INCUMBENT	7090001	0 / 791,530		0 / 791,530
INCUMBENT	7090002	0 / 792,734		0 / 792,734
F-cheap	7090001	1 / 792,140		0 / 792,140
F-cheap	7090002	2 / 795,218		0 / 795,224
composite	7090001	2 / 814,702		0 / 814,726
composite	7090002	0 / 816,225		0 / 816,225

protocol anticipated in terms — a clean run “was not clean by luck; it was a deal partition under which the residue happened not to bite”.

The denominators move between the two columns in 3 of the cells. That is the protocol’s own point restated as measurement: rebuilding agents per deal *changes play*, so the gate certifies a mode no scored cell uses, and the residue is not removed from the shipped one.

13.9 Throughput: the cost claim, checked

The record carried the configuration at “roughly 3.2× the blueprint” and the protocol asked for that to be checked rather than repeated. Every panel cell already carries a throughput figure at a fixed thread count, so the multiple is read off the panel without spending a game on a separate benchmark.

The record understates the cost. The 3.2× the record carried is a figure for **F-cheap**, so the like-for-like comparison is **F-cheap**’s own 4.10×; **FROZEN** costs at least 4.52×. All three tight bounds exceed 3.2×. The whole-panel column is lower only because the expensive panel members put a large common term on both sides of the ratio; it is the most diluted of the three and is printed last for that reason.

14 A fresh adversary search against the frozen configuration

The protocol’s fifth failure condition is that any fitted adversary clears the detection floor against the frozen configuration with a replicated sign: “then v0.7 closed nothing and opened something, and the exploit is the headline”. Eight independent searches were run to find out.

8 of 8 searches produced a weight vector and all 8 were evaluated. **Not one arm beats the frozen configuration.** Every one loses — by between 2.91 and 10.70 points — every sign replicates, and the largest upper bound anywhere is −2.27, which is 3.80 points below the floor. The preregistered claim that no fresh adversary exploits the configuration passes.

Table 17: Cost relative to the blueprint. Every figure is a *lower bound*: the harness reports whole-match throughput, which carries the opponent’s cost on both sides of the ratio, so the measured ratio is strictly below the true one. The bound is tightest against the cheapest opponent. No figure here carries an interval — these are wall-clock timings taken across a battery that ran over two days — and the aggregation is a ratio of medians, which differs from a median of per-bank ratios — the aggregation the phase-5 reduction uses — by at most $0.029\times$ across the six median cells, and not at all on the **random** column that carries the claim.

Arm	vs random (tightest)	Far archetypes (median)	Whole panel (median)
FROZEN	$4.52\times$	$4.72\times$	$2.72\times$
F-cheap	$4.10\times$	$4.36\times$	$2.63\times$
composite	$4.20\times$	$4.51\times$	$2.69\times$

Table 18: Eight independent adversary searches against FROZEN. A positive edge would mean the adversary beats the target; every one is negative. Each arm was fitted on a stream disjoint from the banks it is evaluated on.

ID	Class	Objective	Edge over Frozen	95% CI	Per bank	Sign replicates
Z01	C1 in-class	win	-2.91	$[-3.55, -2.27]$	7090001:-2.76 7090002:-3.06	yes
Z02	C2 extended	win	-3.90	$[-4.52, -3.28]$	7090001:-3.80 7090002:-4.00	yes
Z03	C1	declerr	-10.70	$[-11.32, -10.09]$	7090001:-10.65 7090002:-10.76	yes
Z04	C1	events	-9.64	$[-10.27, -9.01]$	7090001:-9.83 7090002:-9.46	yes
Z05	C1	forced	-7.14	$[-7.76, -6.52]$	7090001:-7.17 7090002:-7.11	yes
Z06	C2 wide sigma	win	-5.25	$[-5.88, -4.62]$	7090001:-4.93 7090002:-5.57	yes
Z07	C1 v05 basin	win	-5.21	$[-5.84, -4.58]$	7090001:-4.85 7090002:-5.58	yes
Z08	C5 white-box	win	-3.56	$[-4.19, -2.92]$	7090001:-3.95 7090002:-3.17	yes

The pass condition and the fail condition are not complements, and both were computed. “Every arm’s upper bound below its floor” and “any arm clears its floor with a replicated sign” leave a band in which the protocol determines no verdict. Had that band been occupied the protocol required the evaluation to stop rather than to resolve it by choosing a reading. The band is *empty* — every upper bound is below its floor — so the pass condition is met outright and no reading had to be chosen. This is recorded because it was a live risk, not because it materialised.

Four objectives, not one. The searches maximise win rate, declaration error, event count and forced-endgame incidence. The declaration-objective arm is read against the declaration-family floor of 2.13 rather than 1.53, because that is its whole objective; its upper bound is far enough below both that the distinction changes no verdict here. Two arms deliberately depart from the template — one with a wide search variance, one seeded in the v0.5 basin rather than the incumbent’s — so that the eight are not eight runs of one search.

14.1 The white-box arm, and its two limits

Zeight is the class the threat model of §8 was written for: an inverter that reads the target’s own deterministic policy and reconstructs the deal posterior from the public transcript. It reaches -3.56 $[-4.19, -2.92]$ — a loss, replicated.

Two limits are stated because this is the cell a reader will weigh most heavily.

First, **18 of its fitted coordinates are inert**. The white-box base derives from the v0.6 agent and reads only that policy’s 37 coordinates, while the search extends its vector to 55 for any base

of this family. The search is therefore over 37 live coordinates plus the inverter the base fixes, and a reader should price it as that rather than as a 55-dimensional search.

Second, **the budget is small**: 8 generations of a 12-member population over 120 deals a candidate. **Zeight** is evidence that this white-box inverter, at this budget, does not exploit the frozen configuration. It is not evidence that no white-box attack can. §17 says so again, because this is the exact place the corpus has previously turned a null result into a win.

14.2 The winner’s curse, visible in the fitting-to-holdout gap

The gap between fitting performance and holdout performance is large and in the expected direction: the in-class arm reached +6.67 points of paired margin in its final generation on its own fitting stream — +8.33 at its best generation — and lands at -2.91 on disjoint banks. That is the arithmetic the disjointness rule exists to expose, and it is why an exploitability number measured on its own fitting stream is not a measurement.

A second sample of the same finding sits in §13.7: the responders fitted for the planted-edge controls, on a different bank and against a handicapped target, also lose to unhandicapped FROZEN in every row. They maximise win rate, as four of these eight arms do, so it is not an independent *method*.

15 What went against the release

The protocol wrote down seven conditions that would mean v0.7 is not an advancement, in advance, because after the numbers are in every one of them has a reading that makes it sound like something else. Two came out against the release. This section is where they are treated as what they are, and it is placed before the corrections and the limitations rather than after them.

15.1 The composite cell: v0.7 does not beat what phase 2 already had

FROZEN **does not beat** the phase-2 composite: **+0.15 points** $[-0.29, +0.59]$ **over 48,000 games**. Both banks are positive, so the sign replicates, but the interval contains zero.

The protocol states the consequence in advance and it is quoted rather than paraphrased: what an earlier phase called “+1.42 $[+0.18, +2.68]$ on one bank” did not replicate, and on this cell the release reduces to the phase-2 composite — “which is a phase-2 result, not a v0.7 one. The report must then say that the v0.7 contribution is the *measurement programme* and the four closed ledger entries, not the policy.”

That is what this paper says.

The arithmetic makes the cell interpretable rather than merely disappointing. FROZEN minus the phase-2 composite is exactly the group `rtie=1 + urgency-off + stall=12` added and `m2=0` removed, and §13.3 measures `stall=12`’s contribution as identically zero and `m2=0`’s leave-one-out drop as zero. The protocol states that group’s training value in advance as +0.78 $[+0.33, +1.22]$. On holdout it is +0.15 $[-0.29, +0.59]$. §15.3 is what the selection-bias check makes of that pair, and the answer is not comfortable.

What this does and does not license. It does not retract §13.2: the frozen configuration *is* a certified advancement over the v0.6 frontier, and **F-cheap** and the phase-2 composite are different objects — the first is a point of the v0.6 frontier, the second is a configuration built during phase 2 of *this* programme. What it licenses is the narrower and more useful statement: **the v0.7 programme produced a real advance over v0.6, and it produced it in phase 2. The architecture**

Table 19: Every quantity for which the protocol states a training figure in advance, against its holdout measurement. Three reproduce or exceed; one falls short.

Quantity	Training	Holdout	Shortfall
B2.4 — <code>rtie</code> + urgency-off + <code>stall</code> , as a group	+0.78	+0.15	−0.63
B6 self-play row, opponent <code>v05</code>	+2.94	+4.49	+1.55
B7 diagonal (self-play)	+4.51	+4.55	+0.04
B7 off-diagonal (cross-play)	+4.48	+4.55	+0.07

work of phase 3, the bake-off of phase 4 and the freeze added no measurable strength on top of it.

15.2 The location test does not replicate

The protocol’s third failure condition is that the lattice cannot locate the gain: if removing every single component in turn costs less than a third of the whole, then the configuration has a number it cannot explain, and the standing rule that “v0.7 must not headline a gain it cannot attribute” binds.

Under the replication rule of §7.2 this must be read per bank, and the two banks disagree:

Bank	Whole	One third	Largest single drop	Verdict
7090001	4.883	1.628	L- <code>r12</code> +2.350	LOCATED
7090003	4.575	1.525	L- <code>r12</code> +1.025	NOT LOCATED

The two banks’ largest drops differ by 1.325 points, against a half-width on that difference of comparable size: the split is real on the point estimates and is *not itself statistically resolved*. Under the rule that a claim whose sign does not replicate is reported as not replicated whatever its pooled interval says, **the pooled verdict is NOT REPLICATED** (no), and the honest reading is that this battery does not settle whether the gain is located by the protocol’s one-third rule.

The sub-additivity claim of §13.3 passes on both banks. It is the one-third location test specifically that splits, and the distinction matters: the components compose predictably, but which one carries the majority of the whole is bank-dependent at this power.

15.3 The selection signature, and where it appears

15 configurations were scored against the deployed policy before the freeze. At the lattice cell size the per-cell standard deviation is 0.3163 points, so the expected maximum of 15 draws under the null that none of them differs from the incumbent is $\sigma\sqrt{2\ln K} = 0.74$ points.

How much of the measured gain would be expected from selection alone: 0.00 points. The holdout banks were never available for selection — not one of the 15 choices could have been made using any deal of them — so the selection term on holdout is zero by construction and not by measurement. That is the strongest structural guarantee in this paper and it is worth stating plainly before the awkward part.

The awkward part is the check’s other direction: whether the holdout estimate is systematically smaller than the training estimate by about that amount.

3 of the four reproduce or exceed their training values. The one that falls short is the composite cell, by 0.63 points against an expected-maximum-under-the-null of 0.74 — **and that cell is precisely the one measuring the group of keys the freeze was selected over.** The shortfall

is the largest of the four and it falls on the cell measuring what the freeze was chosen over, at about the size the null predicts. It is not the only shortfall the protocol’s advance figures produce — the one-seat upgrade against a v0.6 opponent comes in at +2.75 against a predicted +2.88, which §13.4 prints — so the correct statement is that the largest shortfall is where selection happened, not that it is the only one. The training interval for that group was [+0.33, +1.22]; the holdout interval is [−0.29, +0.59].

This is not proof that the group’s training value was entirely selection — one quantity cannot carry that — but it is the pattern the check was written to look for, it is where the check said to look, and it is consistent with §15.1’s reading rather than with an alternative one.

15.4 The seven conditions, answered

#	Condition	Result
1	the frontier cell	not met. +3.33 [+2.88, +3.78], sign replicates
2	the composite cell	MET. +0.15 [−0.29, +0.59]
3	attribution: the location test	DOES NOT REPLICATE. LOCATED on one bank, NOT LOCATED on the other
4	brittleness under partner change or cross-play	not met. Minimum -0.19; cross-play gap +0.01
5	a fresh exploit	not met. Every arm loses; largest upper bound -2.27
6	the gate	not met. All 8 rules pass
7	the controls	not met. Every control behaves as specified

Two of the seven are met or unresolved: item 2 outright, and item 3 by non-replication. The other five are not met. A report that led with the five and buried the two would be describing a different evaluation from the one that ran.

16 Corrections to the record

This project’s convention is that corrections are stated as corrections, in their own section, and not silently patched into the artifacts they correct. The v0.6 study opened a register against the v0.4 and v0.5 record; this cycle adds the entries below. Each names the measurement that forces it.

16.1 Corrections to the v0.4–v0.6 record

The cost of the frontier’s search, wrong in both directions. The corpus recorded the searching configuration at 242× the blueprint, and the v0.6 paper at “three orders of magnitude”. Both figures describe the *unrestricted* configuration, which is not the point of the frontier anything was measured at. The endgame-restricted operating point was re-measured at ~3.2× during this cycle, and that correction was carried into the protocol. **This report corrects the correction:** measured on the shared panel at a fixed thread count, **F-cheap** — the configuration the 3.2× describes — costs at least 4.10× the blueprint, and **FROZEN** at least 4.52× (§13.9). Every figure in that table is a lower bound and none carries an interval. The record has now been wrong in both directions and the reason is the same both times — a ratio quoted without the basis it was computed on.

The corpus’s exploitability figure is a lower bound, and has been read as more. The v0.4 findings document reports that a policy fitted specifically against v0.4 fails to reach parity with it; the same study’s own table records the response reaching 51.19% with an interval spanning parity. The v0.6 study corrected this. It is repeated here because this cycle’s own strongest result — §14,

eight searches, all losing — is a number of exactly the same kind, and the correction is what stops this paper making the same claim.

The declaration-allocation channel is worth zero, not two points. The register ranked it second by priority, at 0.48. §10.1 closes it at exactly zero over ~210,000 recorded declarations, shows its proposed fix is worse than the object it would replace, and shows most of the residual error class is an information limit rather than a defect.

The forced-endgame entry’s own number was wrong. Re-derived from this cycle’s capture, closing the entire accuracy gap is worth 0.030 win-rate points — the register’s figure corrected *upward* by a factor of two, and still a fiftieth of the detection floor. The entry is closed either way.

The detection floor does not buy down with evaluation games. Phase 1 measured 1.68 and predicted the floor would fall as the inverse square root of evaluation games. At exactly four times the games it is 1.53, and the sub-floor rung is still undetected because the excess resolves *to zero* rather than because the interval is wide. Below roughly a point and a half the responder is empty, not unresolved. This report reproduces that on holdout material (§13.7).

16.2 Corrections this cycle makes to its own record

A freeze-verification step could not fire, and its silence was not a pass. The freeze checker reads its source-drift baseline out of the freeze artifact’s own provenance field, and that artifact was rewritten twice after the freeze commit, each rewrite refreshing the baseline to the then-current tree. The check therefore compared the tree against a baseline taken from the tree, and the sources the protocol predicted would be listed as changed no longer differed from it. Measured against the actual freeze commit, five of the 78 sources differ. **What is not weakened is the policy identity:** the round-trip assertions and the mirror digest are computed by *running* the engine, not by comparing file hashes, and all of them round-trip. Because the hash-based check was vacuous it was replaced by an executed one — the engine was rebuilt from the tree as it stood at the end of the battery and the mirror digest recomputed with the fresh binary. It is 5f81f440fc9c272a87e87c05fecc7b74, identical to the frozen artifact and to the binary that played every cell.

A scored cell is not bit-reproducible for a searching arm. Before any holdout material was spent, one training cell was checked to be bit-identical over 5 runs across three thread counts, and it was (yes). **That check was too small.** The panel contains three policies twice, so twenty-four pairs of byte-identical commands were run twice each; one pair disagreed, by one game in 12,000. That is the cross-deal agent residue of §13.8 reaching the *scored* mode, which four hundred deals was not large enough to expose. The magnitude is roughly $76\times$ below a cell half-width and no verdict in this paper turns on it, but the earlier claim should be read as “reproducible at four hundred deals”, not “reproducible”.

The scored-cell count is not the count the artifacts support. That document states 409 scored match cells in three places; the figure is hand-typed in the reduction script rather than computed. Counting rows that carry a match object across every scored battery gives **428**. Nothing turns on it — it is the denominator of an observation about a single action-limit hit — but a corpus whose standing rule is that no number is hand-typed should not carry a hand-typed total, and the generator for this paper now computes it.

The partner-regime table was published without its own intervals. The phase-5 results document prints the partner deltas as bare point estimates with a replication flag. Every underlying cell carries a deal-clustered interval, and the delta arithmetic is fixed by the protocol, so the intervals are computable and were simply compressed out of a markdown table. Table 9 restores them. No verdict changes; a reader’s ability to judge the verdict does.

The cross-play gap was printed under the reversed sign convention. The phase-5 document

reports the gap as -0.01, which is diagonal minus off-diagonal. The claim it settles asks whether the *off*-diagonal collapses, so the informative direction is off-diagonal minus diagonal, which is +0.01. *Neither sign is resolved*: the magnitude is 0.01 points against a per-cell half-width of 0.63 and a 1.50-point threshold, so no verdict moves and this paper reads no lift out of it either. What the correction is worth is only this: the sign as printed *implies* a small collapse, and the data support no direction at all.

16.3 A note on comparability

Two things about this cycle's evidence standard are not the corpus's, and a reader comparing across versions should have them. Every claim here is measured on material sealed before the configuration existed, under a protocol committed before any of it was played, with a stated failure condition — which no previous study in this corpus did. And every claim is replicated across two disjoint banks with the sign rule of §7.2 applied, which is why §15.2 reports a non-replication that a pooled-interval convention would have reported as a pass.

17 Threats to validity, and the weaknesses that remain

The release adds nothing measurable to what phase 2 already had. This is the sharpest limitation and it is measured, not conjectured: +0.15 points $[-0.29, +0.59]$ against the phase-2 composite (§15.1). Everything else in this paper describes a configuration whose strength advantage over the v0.6 frontier is real and was found in phase 2 by an unfitted coordinate sweep.

The gain is located in one coordinate, and the location does not replicate. r12=25 carries it, with a leave-one-out drop of +1.69 $[+0.80, +2.57]$. But the protocol's own one-third location test is LOCATED on one bank and NOT LOCATED on the other (§15.2). A configuration whose gain is concentrated in a single coordinate is also a configuration with a single point of failure, and this battery does not resolve how concentrated it is.

The worst-case improvement and the minimax regret point in opposite directions. FROZEN has a far better worst cell than INCUMBENT but not the best of the four — the phase-2 composite is ahead on both readings — and FROZEN is 3rd of four on minimax regret (§13.1). Its worst cell does not even replicate in sign (+1.62 / -1.75, at different members). The regret difference is bought against far archetypes that every arm already beats by 24.72 to 50.00 points, which is a reason to weigh it less — but it is a reason, not a dismissal, and a reader who cares about regret over this panel should prefer **F-cheap**.

Exploitability is a lower bound within a searched class. §14 fits eight responders from named classes at a stated budget. A stronger response outside those classes, or the same classes at a larger budget, would give a larger number. A response that fails to reach the target proves nothing about the target; it constrains the search. The white-box arm in particular runs 37 live coordinates at 8 generations of 12 (§14.1). This is the reading error the corpus has made before and it is named here rather than risked.

The opponent panel is a modelling choice and its deception archetypes are hand-built. The withholder, the silent and the feint are stylised implementations of manoeuvres the project owner reported from live play. They are not people. A gain against one of them is evidence about

robustness to a class of behaviour, not a measurement of play against a person. The partner columns have the same limitation from the other side: a “foreign partner” here is a weaker or deceptive bot.

The advantage transfers across partners less well than the incumbent’s did, on the ratio the protocol chose in advance. 0.313 against 0.380 (§13.4). The collapse test passes and the absolute changed-partner median is much larger than the incumbent’s, but the pre-committed sentence is the one this paper uses, and it is not “it transfers”.

A named engine defect is present in every number here. Agents are reused across deals within a thread and some per-agent state survives reset, reachable only under the search. The carrier is named — `V06Agent::roll` — the member is not, the gate condition removes it by rebuilding agents per deal, and *that mode is not the shipped mode*. §13.8 shows the audited denominators moving between the two conditions, which is the defect made visible rather than argued about. Every strength number in this paper is measured under it.

Four of the seven sealed banks were digest-verified and never played as deal banks. The fitting driver derives its own per-generation deal stream from its seed rather than constructing the named bank’s deals, so 7090004, 7090005 and 7091002 are verified and then unused as deals. 7091001 is unused for a different reason: the sealed panel members were scored on the two primary banks with the rest of the panel, so that minimax regret is computed over one shared pair. What the protocol relies on survives — the fitting streams are mutually disjoint and disjoint from the evaluation banks — but the material accounting here should be read as “three evaluation banks played (7090001 and 7090002 for most cells, 7090003 for the lattice and the dialects) plus a derived fitting stream”, not “seven banks played”.

One dialect row measures nothing, and the residual it feeds is correspondingly weaker. `-maxasks=360` is bit-identical to the default, so the legacy residual is legacy minus two effective components rather than three, and its interval is $9.6\times$ the size of the residual itself (§13.6). The data are equally consistent with the forced-endgame willingness ladder contributing nothing at all.

The negative results are null at this power and against this panel. Four architectures are reported as worth nothing (§10). Each was measured paired, on two disjoint banks, against a multi-opponent panel. That resolves effects of roughly a point; it does not resolve effects of a quarter of a point. What this cycle establishes is that none of the architectures on the inherited register is large — not that the sum of many small ones is zero.

Cross-play cannot see a convention inherited from the shared ancestor. The three independently-trained runs of §13.5 all start from or near the v0.6 vector. A convention that all three inherited from that common ancestor would appear on the diagonal and the off-diagonal alike and would be invisible to this test. What the test rules out is a convention formed *during* each run’s own fit.

Termination remains empirical. No configuration here hits the action cap in any scored cell but one — composite against `SEALED:S-ask-1`, one game in 12,000 — and the stall rung’s separation between ordinary play and a deadlock is a measurement over 800 mirror games rather than a

theorem. What is not proved is that the stall rule plus the stopping rule terminates on every reachable position.

18 What v1.0 would have to show

Two sentences a future release might want to write are “FishBot v1.0 is near-optimal” and “FishBot v1.0 is the strongest known Fish-playing team”. Neither is supportable by anything in this corpus today, and neither is close. This section states what evidence each would require, so that the gap is a list of experiments rather than a matter of taste. Nothing here has been measured; every item is open.

18.1 For “near-optimal”

Near-optimality is a claim about distance to a benchmark, so the first requirement is a benchmark. The corpus has none.

1. **An upper bound on exploitability, not a lower one.** Every exploitability figure in this corpus, including all eight arms of §14, is a lower bound produced by a search: it says what *this* search found, and computing the true value is NP-hard even white-box. A near-optimality claim needs a bound in the other direction — an argument or a computation showing that no adversary in a stated class can exceed some ε . No such bound exists here for any class, and producing one for even a restricted class would be the single most valuable result the next cycle could return.
2. **A team analogue of local best response, and the rollout policy it needs.** The poker line’s standard exploitability probe rests on a cheap, correct-enough passive continuation — “what if nobody does anything clever from here”. Fish has no passive continuation: a player must ask or declare. Until a defensible rollout policy for that role is named and calibrated, the probe cannot be ported, and the literature’s own warning that choosing a value function is the hard part applies directly.
3. **An adversary that is genuinely ex-ante correlated.** The threat model names three correlation regimes and makes the correlated one the headline. What has been built and measured is the synchronized regime, plus one correlated fit that reached +0.70 against a class floor of 2.31 — i.e. nothing. Whether a genuinely correlated three-seat responder is buildable at all for a game this size is open, and the honest position is that the headline regime has fallen back to a weaker one.
4. **A detection floor low enough for the claim.** The instrument’s floor is 1.53 points and §16.1 records that it does not buy down with evaluation games. “Near-optimal” is not a statement compatible with an instrument that cannot distinguish a policy from one a point and a half weaker. Lowering the floor is not an evaluation-power problem — it is a search-power problem, and phase 2 established that by measurement.
5. **The sum of small effects, addressed.** This report and its predecessor both conclude that no *individual* mechanism on the register is large. Neither addresses whether a dozen quarter-point mechanisms exist and compound. Resolving *one* point on the scoreboard needs roughly 9,604 games a cell, and a quarter of a point sixteen times that; the per-decision estimator of §10.3 resolves the same one-point effect in 284–560. That estimator is confirmed and unused, because fitting against it fails. Using it as an *estimator* while fitting against the scoreboard is the obvious unattempted experiment.

18.2 For “the strongest known Fish-playing team”

This is a comparative claim and it is currently unfalsifiable, because the comparison class has one member.

1. **An opponent this project did not write.** Every arm, archetype and adversary in this corpus was built inside it. The panel’s deception archetypes are stylised implementations of manoeuvres one person reported from live play. A claim to be strongest known requires at least one independently authored opponent, and ideally a published rule dialect other people can implement against.
2. **Play against people, measured.** The corpus has no human-play measurement of any kind. What it has is a partner-regime table in which “a foreign partner” is another bot (§13.4). The project owner’s standing decision — never headline the self-play configuration as the one a person will meet — exists precisely because the substitute is not the thing. A strongest-known claim needs a duplicate-style human battery with enough deals to resolve a point, and the six-seat duplicate design for it has not been settled.
3. **A stated, published dialect.** Fish is played in many house dialects, and §13.6 shows the arm ordering is stable across eight of them — which is evidence the claim could survive dialect variation, and also evidence that “Fish” names a family rather than a game. Any comparative claim has to name the dialect it holds in.
4. **Somebody else’s evaluation.** Every number in this paper was produced by one engine, written by this project, audited by tests this project wrote. The strongest available substitute for external replication was used — material sealed before the configuration existed, digests computed by a second build, a protocol committed in advance — and it is not the same thing.

18.3 The next cycle’s first three jobs

Stated concretely, because a barrier report should hand over work rather than a mood.

1. **Fix the cross-deal agent residue, and re-measure everything the fix moves.** The carrier is named (`V06Agent::roll`); the member is not. The repair changes the play of every searching configuration by about one deal in 800. It is the first maintenance item after this freeze, and it must precede the next freeze rather than follow it.
2. **Test whether the small effects compound.** Per-decision estimation, per-game objective, on a lattice of mechanisms each individually below the floor. This is the one hypothesis both this cycle and the last explicitly declined to test, and both said so.
3. **Build an upper bound for one restricted adversary class.** Even a loose bound over a small class would convert every exploitability sentence in this corpus from “we did not find one” to “there is not one larger than ε ”, which is a different kind of claim and the only kind that supports the word near-optimal.

19 Conclusion

The v0.7 programme set out to find whether the v0.6 policy class is flat or merely badly measured. It built an instrument, characterised what that instrument can see, generated adversaries in the open, built five candidate architectures, froze one configuration, and then measured it against material sealed before the configuration existed under a protocol that said in advance what would count as failure.

The class is flat, and the evidence for that is the shape of what survived.

One configuration is a certified advancement over the v0.6 frontier: +3.33 points [+2.88, +3.78] against **F-cheap** over 48,000 games of sealed material, replicating on both banks, gate-clean, and with a worst cell on the shared 31-member panel (-0.04) far better than the incumbent's -4.24. Nothing built to exploit it succeeds — eight independent searches, the white-box inverter included, every one losing, the largest upper bound -2.27. It is not a private convention: a single seat's upgrade is worth +1.93 points of the +4.49-point three-seat gain, partner substitution leaves a minimum of -0.19, and three independently-trained runs cross-play within 0.01 points of their own diagonal.

And it adds nothing measurable to the configuration phase 2 already had (+0.15 [-0.29, +0.59]). The gain that exists was found by an unfitted coordinate sweep, not by architecture work. The three phases of architecture work that followed produced four kills and one configuration change. The one quantity that falls short of its training value is the group of keys the freeze was selected over, by 0.63 points against an expected maximum under the null of 0.74. And the protocol's own test for whether the gain is located does not replicate across banks.

So the contribution of this cycle is the closed directions and the named barrier, and the protocol said in advance that this would be a legitimate deliverable rather than a failure to apologise for.

The closed directions. The declaration-allocation channel — the register's second-ranked entry — is worth exactly zero, its named mechanism shown not to exist over ~210,000 recorded declarations, its proposed fix shown to be worse than the object it would replace, and most of its residual error class shown to be an information limit. A leaf evaluator fitted $74\times$ better in the only statistic its consumer can use buys nothing, because that consumer is nearly leaf-invariant. Fitting against any self-oriented per-decision objective produces a worse policy while moving its own proxy the right way, because the two per-decision channels trade against each other. Nothing learned survives a free random tie-break inside the exchangeable tie group, whose selection signal is ~84% winner's curse. Two further directions closed on arithmetic before they were built. None of these is a null anyone need re-run.

The barrier. It is not that the mechanisms were badly built — three of the four moved their own proxies in the intended direction and one improved its fit statistic by $74\times$. It is that *this policy class converts mechanism into win rate at a rate below the instrument's floor*, and the floor does not buy down with evaluation games. The one thing that did convert was a single coordinate found by sweeping, not by designing. A programme that wants to move this policy further has to change one of the two terms: build a search that can find effects below 1.53 points, or find out whether a dozen quarter-point mechanisms compound. This cycle did neither, and the last one did not either, and both said so.

What should not be claimed. No policy in this corpus is near-optimal, and none is known to be the strongest Fish-playing team, because the comparison class has one member and every exploitability number in the corpus is a lower bound from a search rather than a bound on what a search could find. §18 lists what would have to be measured before either sentence could be written. It is a list of experiments, not of opinions, and none of them has been run.

References

- [1] J. McLeod, “Literature,” *Pagat*, 2006–2025, last updated 1 July 2026.
<https://www.pagat.com/quartet/literature.html>
- [2] M. Develin, “Canadian Fish,” chapter 9 of *Card Games*, author’s online manual.
<https://web.archive.org/web/20170720075756/http://bantha.org/~develin/cardgames.html#ch9>
- [3] Deposit Genius, “Literature Game Strategy (Canadian Fish),” online article.
<https://depositgenius.com/literature-strategy-canadian-fish/>
- [4] Canadian Fish Project, “Strategy notes,” 2026.
<https://canadian-fish.vercel.app/strategy>
- [5] Wikipedia, “Literature (card game),” online encyclopedia entry.
[https://en.wikipedia.org/wiki/Literature_\(card_game\)](https://en.wikipedia.org/wiki/Literature_(card_game))
- [6] N. Somani, “literature: card game implementation and learning bot,” GitHub repository, MIT licence, Python, 2018–.
<https://github.com/neelsomani/literature>
- [7] N. Somani, “literature-server,” GitHub repository, with a live deployment at <https://literature.neelsomani.com/>
<https://github.com/neelsomani/literature-server>
- [8] “Literature: Fish Card Game” (`com.cards.game.literature`), Google Play store listing, closed source.
<https://play.google.com/store/apps/details?id=com.cards.game.literature>
- [9] D. Zha, K.-H. Lai, S. Huang, Y. Cao, K. Reddy, J. Vargas, R. Wei, J. Guo, and X. Hu, “RLCard: A Toolkit for Reinforcement Learning in Card Games,” arXiv:1910.04376, 2019.
- [10] M. Goadrich, A. Morenville, and É. Piette, “Valet: A Standardized Testbed of Traditional Imperfect-Information Card Games,” arXiv:2603.03252, 2026. 21 games in the RECYCLE description language; whether any of them is a team game is not stated on the abstract page and the full text was not fetched.
<https://arxiv.org/abs/2603.03252>
- [11] D. Levy, “The Million Pound Bridge Program,” *Heuristic Programming in Artificial Intelligence: The First Computer Olympiad*, Ellis Horwood, 1989.
- [12] M. L. Ginsberg, “GIB: Imperfect Information in a Computationally Challenging Game,” *Journal of Artificial Intelligence Research* 14:303–358, 2001.
- [13] I. Frank and D. Basin, “Search in Games with Incomplete Information: A Case Study Using Bridge Card Play,” *Artificial Intelligence* 100(1–2):87–123, 1998.
- [14] I. Frank, D. Basin, and H. Matsubara, “Finding Optimal Strategies for Imperfect Information Games,” *AAAI-98*, pp. 500–507, 1998.
- [15] J. Long, N. R. Sturtevant, M. Buro, and T. Furtak, “Understanding the Success of Perfect Information Monte Carlo Sampling in Game Tree Search,” *AAAI-10*, pp. 134–140, 2010.

- [16] P. I. Cowling, E. J. Powley, and D. Whitehouse, “Information Set Monte Carlo Tree Search,” *IEEE Transactions on Computational Intelligence and AI in Games* 4(2):120–143, 2012.
- [17] J. Goodman, “Re-determinizing Information Set Monte Carlo Tree Search in Hanabi,” arXiv:1902.06075, 2019.
- [18] T. Cazenave and V. Ventos, “The $\alpha\mu$ Search Algorithm for the Game of Bridge,” arXiv:1911.07960, 2019.
<https://arxiv.org/abs/1911.07960>
- [19] S. Edelkamp, “Knowledge-Based Paranoia Search in Trick-Taking,” arXiv:2104.05423, 2021. Skat; over 1,000 points on the extended Seeger scale. The runtime is not stated on the abstract page, so no compute cost is attributed to this method in this report.
<https://arxiv.org/abs/2104.05423>
- [20] M. Buro, J. R. Long, T. Furtak, and N. R. Sturtevant, “Improving State Evaluation, Inference, and Search in Trick-Based Card Games,” *IJCAI-09*, pp. 1407–1413, 2009.
- [21] C. Solinas, D. Rebstock, and M. Buro, “Improving Search with Supervised Learning in Trick-Based Card Games,” *AAAI-19*, 2019; arXiv:1903.09604.
- [22] D. Rebstock, C. Solinas, M. Buro, and N. R. Sturtevant, “Policy Based Inference in Trick-Taking Card Games,” *IEEE Conference on Games 2019*, 2019; arXiv:1905.10911.
- [23] C. Solinas, D. Rebstock, N. R. Sturtevant, and M. Buro, “History Filtering in Imperfect Information Games: Algorithms and Complexity,” *NeurIPS 2023*, 2023; arXiv:2311.14651.
- [24] N. Brown, T. Sandholm, and B. Amos, “Depth-Limited Solving for Imperfect-Information Games,” *NeurIPS 2018*, 2018; arXiv:1805.08195. Master-level heads-up no-limit hold’em on a 4-core CPU and 16 GB, via multi-valued states. The soundness argument is for two-player zero-sum games and is not inherited by the team game studied here.
<https://arxiv.org/abs/1805.08195>
- [25] B. H. Zhang and T. Sandholm, “Subgame Solving without Common Knowledge,” *NeurIPS 2021*, 2021; arXiv:2106.06068.
<https://proceedings.neurips.cc/paper/2021/hash/c96c08f8bb7960e11a1239352a479053-Abstract.html>
- [26] B. H. Zhang, L. Carminati, F. Cacciamani, G. Farina, P. Olivieri, N. Gatti, and T. Sandholm, “Subgame Solving in Adversarial Team Games,” *NeurIPS 2022*, 2022. Cited from the proceedings abstract page; the mirrored PDF did not decode.
https://proceedings.neurips.cc/paper_files/paper/2022/hash/aa5f5e6eb6f613ec412f1d948dfa21a5-Abstract-Conference.html
- [27] S. Sokota, G. Farina, D. J. Wu, H. Hu, K. A. Wang, J. Z. Kolter, and N. Brown, “The Update-Equivalence Framework for Decision-Time Planning,” *ICLR 2024*, 2024; arXiv:2304.13138. Mirror-descent search at two orders of magnitude less search time than public-belief-state search in Hanabi.
<https://openreview.net/forum?id=JXGph215fL>
- [28] A. Lerer, H. Hu, J. Foerster, and N. Brown, “Improving Policies via Search in Cooperative Partially Observable Games,” *AAAI 2020*, 2020; arXiv:1912.02318. SPARTA; Hanabi 24.08 to

24.61 at test time.

<https://ojs.aaai.org/index.php/AAAI/article/view/6208>

- [29] H. Hu, D. J. Wu, A. Lerer, J. Foerster, and N. Brown, “Learned Belief Search: Efficiently Improving Policies in Partially Observable Settings,” arXiv:2106.09086, 2021.
<https://arxiv.org/abs/2106.09086>
- [30] A. Fickinger, H. Hu, B. Amos, S. Russell, and N. Brown, “Scalable Online Planning via Reinforcement Learning Fine-Tuning,” *NeurIPS 2021*, 2021; arXiv:2109.15316.
<https://arxiv.org/abs/2109.15316>
- [31] D. Milec, O. Kubíček, and V. Lisý, “Continual Depth-limited Responses for Computing Counter-strategies in Sequential Games,” arXiv:2112.12594, 2021. Robust-response methods do not compose with limited-look-ahead solving off the shelf.
<https://arxiv.org/abs/2112.12594>
- [32] D. Milec, V. Kovařík, and V. Lisý, “Adapting Beyond the Depth Limit: Counter Strategies in Large Imperfect Information Games,” arXiv:2501.10464, 2025.
<https://arxiv.org/abs/2501.10464>
- [33] O. Kubíček and V. Lisý, “Look-ahead Reasoning with a Learned Model in Imperfect Information Games,” arXiv:2510.05048, 2025; ICLR 2026 per the arXiv PDF header.
<https://arxiv.org/abs/2510.05048>
- [34] O. Kubíček, V. Lisý, and T. Sandholm, “Equilibrium Refinements Improve Subgame Solving in Imperfect-Information Games,” arXiv:2601.17131, 2026.
<https://arxiv.org/abs/2601.17131>
- [35] N. Brown, A. Bakhtin, A. Lerer, and Q. Gong, “Combining Deep Reinforcement Learning and Search for Imperfect-Information Games,” *NeurIPS 2020*, 2020; arXiv:2007.13544.
<https://arxiv.org/abs/2007.13544>
- [36] M. Schmid, M. Moravčík, N. Burch, R. Kadlec, J. Davidson, K. Waugh, N. Bard, F. Timbers, M. Lanctot, G. Z. Holland, E. Davoodi, A. Christianson, and M. Bowling, “Student of Games: A Unified Learning Algorithm for Both Perfect and Imperfect Information Games,” *Science Advances* 9(46), 2023; arXiv:2112.03178. Circulated 2021–2022 as “Player of Games.”
- [37] M. Moravčík, M. Schmid, N. Burch, V. Lisý, D. Morrill, N. Bard, T. Davis, K. Waugh, M. Johanson, and M. Bowling, “DeepStack: Expert-Level Artificial Intelligence in Heads-Up No-Limit Poker,” *Science* 356(6337):508–513, 2017.
- [38] V. Lisý and M. Bowling, “Equilibrium Approximation Quality of Current No-Limit Poker Bots,” arXiv:1612.07547, 2017. Local Best Response; every ACPC 2016 entrant is more than 3,180 mBB/h exploitable.
<https://arxiv.org/abs/1612.07547>
- [39] N. Bard, J. N. Foerster, S. Chandar, N. Burch, M. Lanctot, H. F. Song, E. Parisotto, V. Dumoulin, S. Moitra, E. Hughes, I. Dunning, S. Mourad, H. Larochelle, M. G. Bellemare, and M. Bowling, “The Hanabi Challenge: A New Frontier for AI Research,” *Artificial Intelligence* 280, 2020; arXiv:1902.00506.

- [40] J. N. Foerster, H. F. Song, E. Hughes, N. Burch, I. Dunning, S. Whiteson, M. Botvinick, and M. Bowling, “Bayesian Action Decoder for Deep Multi-Agent Reinforcement Learning,” *ICML 2019*, 2019; arXiv:1811.01458.
<https://arxiv.org/abs/1811.01458>
- [41] H. Hu and J. N. Foerster, “Simplified Action Decoder for Deep Multi-Agent Reinforcement Learning,” *ICLR 2020*, 2020; arXiv:1912.02288.
- [42] H. Hu, A. Lerer, A. Peysakhovich, and J. N. Foerster, “‘Other-Play’ for Zero-Shot Coordination,” *ICML 2020*, 2020; arXiv:2003.02979.
- [43] H. Hu, A. Lerer, B. Cui, L. Pineda, N. Brown, and J. Foerster, “Off-Belief Learning,” *ICML 2021*, 2021; arXiv:2103.04000.
<https://arxiv.org/abs/2103.04000>
- [44] S. Sokota, E. Lockhart, F. Timbers, E. Davoodi, R. D’Orazio, N. Burch, M. Schmid, M. Bowling, and M. Lanctot, “Solving Common-Payoff Games with Approximate Policy Iteration,” *AAAI 2021*, 2021; arXiv:2101.04237.
<https://arxiv.org/abs/2101.04237>
- [45] H. Hu, S. Sokota, D. J. Wu, A. Bakhtin, A. Lupu, B. Cui, and J. N. Foerster, “Self-Explaining Deviations for Coordination,” *NeurIPS 2022*, 2022; arXiv:2207.12322. IMPROVISED; the first algorithmic Hanabi finesse plays.
https://proceedings.neurips.cc/paper_files/paper/2022/hash/faa6276ea12d7afeb3e42b210c86f688-Abstract-Conference.html
- [46] C. Cox, J. De Silva, P. DeOrsey, F. H. J. Kenter, T. Retter, and J. Tobin, “How to Make the Perfect Fireworks Display: Two Strategies for Hanabi,” *Mathematics Magazine* 88(5):323–336, 2015.
- [47] Y. Tian, Q. Gong, and T. Jiang, “Joint Policy Search for Multi-Agent Collaboration with Imperfect Information,” *NeurIPS 2020*, 2020; arXiv:2008.06495. Contract Bridge: +0.63 IMPs/board against WBridge5.
<https://arxiv.org/abs/2008.06495>
- [48] A. P. Jacob, D. J. Wu, G. Farina, A. Lerer, H. Hu, A. Bakhtin, J. Andreas, and N. Brown, “Modeling Strong and Human-Like Gameplay with KL-Regularized Search,” *ICML 2022*, 2022; arXiv:2112.07544.
<https://arxiv.org/abs/2112.07544>
- [49] H. Hu, D. J. Wu, A. Lerer, J. Foerster, and N. Brown, “Human-AI Coordination via Human-Regularized Search and Learning,” arXiv:2210.05125, 2022.
<https://arxiv.org/abs/2210.05125>
- [50] Meta Fundamental AI Research Diplomacy Team (FAIR), A. Bakhtin, N. Brown, E. Dinan, G. Farina, C. Flaherty, D. Fried, A. Goff, J. Gray, H. Hu, and others, “Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning,” *Science* 378(6624):1067–1074, 2022.
<https://www.science.org/doi/10.1126/science.ade9097>
- [51] T. Dizdarević and others, “Ad-Hoc Human-AI Coordination Challenge,” *ICML 2025*, 2025; arXiv:2506.21490.
<https://proceedings.mlr.press/v267/dizdarevic25a.html>

- [52] A. Nayyar, A. Mahajan, and D. Teneketzis, “Decentralized Stochastic Control with Partial History Sharing: A Common Information Approach,” *IEEE Transactions on Automatic Control* 58(7):1644–1658, 2013; arXiv:1209.1695.
<https://arxiv.org/abs/1209.1695>
- [53] A. Celli and N. Gatti, “Computational Results for Extensive-Form Adversarial Team Games,” *AAAI 2018*, 2018; arXiv:1711.06930.
- [54] G. Farina, A. Celli, N. Gatti, and T. Sandholm, “Connecting Optimal Ex-Ante Collusion in Teams to Extensive-Form Correlation: Faster Algorithms and Positive Complexity Results,” *ICML 2021*, PMLR 139:3164–3173, 2021.
- [55] B. H. Zhang, G. Farina, A. Celli, and T. Sandholm, “Team Belief DAG: Generalizing the Sequence Form to Team Games for Fast Computation of Correlated Team Max-Min Equilibria via Regret Minimization,” *ICML 2023*, 2023; arXiv:2202.00789.
- [56] L. Carminati, F. Cacciamani, M. Ciccone, and N. Gatti, “A Marriage between Adversarial Team Games and 2-Player Games: Enabling Abstractions, No-Regret Learning and Subgame Solving,” *ICML 2022*, 2022; arXiv:2206.09161.
- [57] L. Carminati, B. H. Zhang, G. Farina, N. Gatti, and T. Sandholm, “Hidden-Role Games: Equilibrium Concepts and Computation,” *ACM Conference on Economics and Computation (EC) 2024*, 2024; arXiv:2308.16017.
<https://arxiv.org/abs/2308.16017>
- [58] H. J. Ryser, *Combinatorial Mathematics*, Carus Mathematical Monographs 14, Mathematical Association of America, 1963.
- [59] D. G. Glynn, “The Permanent of a Square Matrix,” *European Journal of Combinatorics* 31(7):1887–1891, 2010.
- [60] M. Jerrum, A. Sinclair, and E. Vigoda, “A Polynomial-Time Approximation Algorithm for the Permanent of a Matrix with Non-Negative Entries,” *Journal of the ACM* 51(4):671–697, 2004.
- [61] N. Linial, A. Samorodnitsky, and A. Wigderson, “A Deterministic Strongly Polynomial Algorithm for Matrix Scaling and Approximate Permanents,” *Combinatorica* 20(4):545–568, 2000.
- [62] Y. Chen, P. Diaconis, S. P. Holmes, and J. S. Liu, “Sequential Monte Carlo Methods for Statistical Analysis of Tables,” *Journal of the American Statistical Association* 100(469):109–120, 2005.
- [63] N. Bard, J. Hawkin, J. Rubin, and M. Zinkevich, “The Annual Computer Poker Competition,” *AI Magazine* 34(2):112–118, 2013; and the competition rules describing duplicate matches and common seeds.
- [64] M. Zinkevich, M. Bowling, N. Bard, M. Kan, and D. Billings, “Optimal Unbiased Estimators for Evaluating Agent Performance,” *AAAI-06*, pp. 573–578, 2006.
- [65] N. Burch, M. Schmid, M. Moravčík, D. Morrill, and M. Bowling, “AIVAT: A New Variance Reduction Technique for Agent Evaluation in Imperfect Information Games,” *AAAI-18*, 2018; arXiv:1612.06915.

- [66] K. Waugh, D. Schnizlein, M. Bowling, and D. Szafron, “Abstraction Pathologies in Extensive Games,” *AAMAS 2009*, pp. 781–788, 2009.
- [67] D. Balduzzi, K. Tuyls, J. Pérolat, and T. Graepel, “Re-evaluating Evaluation,” *NeurIPS 2018*, 2018; arXiv:1806.02643.
- [68] R. Coulom, “Whole-History Rating: A Bayesian Rating System for Players of Time-Varying Strength,” *Computers and Games 2008*, 2008.
- [69] R. Herbrich, T. Minka, and T. Graepel, “TrueSkill™: A Bayesian Skill Rating System,” *Advances in Neural Information Processing Systems* 19, 2006.
- [70] M. Van den Bergh, “Comments on Normalized Elo,” Fishtest technical note; and Stockfish contributors, “Statistical Methods and Algorithms in Fishtest.”
https://cantate.be/Fishtest/normalized_elo_practical.pdf
- [71] N. Kelidari, M. Haghi, and M. Salmani, “A Gold-Standard Study of What Makes a Lightweight Game-Playing Agent Strong,” arXiv:2607.06854, 2026. Gin Rummy and Leduc; concludes that the binding constraint is information, not capacity.
<https://arxiv.org/abs/2607.06854>

A The rules dialect, clause by clause

Every contested rule choice of §2.2 is an engine switch, so its effect is measurable rather than assumed. This appendix lists each choice against the switch that changes it, so that a reader can re-implement the dialect or vary it.

Clause	Switch	Setting studied
Deck and half-suits	<code>-sets=8</code>	54 cards in nine half-suits; <code>-sets=8</code> plays the 48-card, eight-half-suit variant
Out-of-turn declaration	<code>-no-out-of-turn</code>	permitted; the switch forbids it
Cardless declaration	<code>-no-cardless-declare</code>	permitted; the switch forbids it
Simultaneous declarers	<code>-arb=high</code> , <code>-arb=turn</code>	lowest seat index first; the switches arbitrate by stated confidence and by turn order
Misdeclaration	—	the half-suit is awarded to the opposing team
Forced endgame	—	eight-level willingness ladder when one whole team is cardless
Action cap	<code>-maxasks=N</code>	400 asks; residue adjudicated by physical majority
Legacy bundle	<code>-legacy</code>	the full v0.3-era dialect, bundling several of the above

§13.6 measures the paper’s headline comparison under 8 rows of this table. Every row keeps its sign and every row replicates; the largest excursion from the default row is +0.78 points, on `arb=turn`.

Two clauses on which an earlier report in this corpus is wrong, corrected here.

1. **An undeclared half-suit does not score nothing.** An earlier report justified an unconditional forcing rule on the premise that an unclaimed half-suit is worthless. Under the dialect actually evaluated, residue at the action cap is awarded by physical majority, and on a wrong post-horizon declaration the declaring team holds a mean 4.08 of the six cards. The rule converts a probable award into a certain concession.
2. **Restarting the forced-endgame ladder sharpens nothing.** The argument is that restarting the willingness ladder after each declaration lets early, safer declarations resolve cards and so sharpen later ones. It is sound and inoperative: on every occasion a team went cardless with live half-suits — 566 of them at seed 31 — exactly 1 half-suit was live, so there is no ordering to exploit.

B Configuration grammar and parameters

The grammar

A configuration is named by a single string, and every policy in this paper — every arm, every panel member, every fitted adversary — is one of these strings. The grammar is

```
base [ :key=value ( ,key=value )* ]
```

where `base` names a policy family (v03 ... v07, a scripted archetype such as `withholder`, or an instrumented variant such as `v07i` for the white-box inverter), and each `key=value` sets one option on it. Two properties of the grammar are load-bearing for the evaluation:

- **It is total.** Any configuration this paper measures can be written as one string and reconstructed from it, which is what makes an ablation lattice mechanical rather than a set of hand-edited builds.
- **An explicit key beats a bulk vector.** A spec may carry `allparams=w1|w2|...`, the entire parameter vector; explicit keys written alongside it are re-applied afterwards. Before this was fixed, a sentinel such as `pool=-1` was silently clamped away by the bulk vector, and three independently fitted configurations carried machinery their spec said was switched off.

The frozen configuration

`v07:r12=25,rtie=1,pool=-1,oppfloor=-1,force=1000000,askfloor=-1,stall=12,s1=1,det=12,cand=4,kappa=2.5,rbelief=indep,depth=12,maxq=26`

Key	Value	What it controls
<code>r12</code>	25	half-suit contestation weight (<code>oppCertDonate</code>)
<code>rtie</code>	1	tie-break by a hash of the public event stream, not sort order
<code>pool</code>	-1	urgency pooling threshold; -1 disables
<code>oppfloor</code>	-1	urgency opponent-ownership floor; -1 disables
<code>force</code>	1000000	urgency forcing horizon in events; 10 ⁶ disables
<code>askfloor</code>	-1	urgency ask floor; -1 disables
<code>stall</code>	12	escalate after K public events with no change in this seat's certificate hash
<code>s1</code>	1	enable the endgame-truncated determinized search
<code>det</code>	12	determinizations sampled per searched decision
<code>cand</code>	4	candidate actions carried into the search
<code>kappa</code>	2.5	shrinkage of the paired lower-confidence-bound deviation rule
<code>rbelief</code>	<code>indep</code>	belief model used to reconstruct continuation players
<code>depth</code>	12	search depth limit in plies
<code>maxq</code>	26	search runs only when at most this many cards remain unresolved

Three of these are switches that cannot be expressed as coordinates of the parameter vector at all — `askfloor`, `pool`, `oppfloor` — which is why the freeze artifact records the option map separately from the vector and why the round-trip check of §11 has to execute rather than compare strings.

The parameter vector

Block	Coordinates
<code>nfeat</code>	20
<code>v05knobs</code>	14
<code>v06askTerms</code>	3
<code>v07responder</code>	18
<code>total</code>	55

the flat ‘allparams’ vector: NFEAT ask weights, the fourteen v0.5 knobs, the three v0.6 ask terms, then the eighteen v0.7 responder coordinates (`v07_responder.hpp r7Name`)

The first 37 coordinates are inherited unchanged from the previous release’s fit, whose provenance is recorded in the freeze artifact as `fitC.jsonl [final weights record] sha256:be4a4c31d9153208 gens=14 obj=minimaxregret paired=1 panel=v05+v03+withholder+fe....` No fit of any kind was applied to the frozen configuration in this cycle (§11.2).

Coordinates known to be inert

Not every fitted coordinate does anything, and a paper that presented all 55 as tuned quantities would be overstating the search. Known inert coordinates in this family:

- two v0.5 knobs unreachable under the shipped declaration branch — setting them produces a bit-identical transcript;
- three v0.5 ask terms belonging to a re-scoring pass that the shipped configuration does not run;
- `m2`, whose leave-one-out drop is exactly zero and which was dropped from the freeze for that reason (§13.3);
- `stall`, which is armed and never fires: its mirror digest is byte-identical to the same configuration without it (§11.1);
- and, in the white-box adversary class specifically, 18 of the 55 fitted coordinates, because that base reads only the 37-coordinate v0.6 vector (§14.1).

C The preregistered thresholds, verbatim

Reproduced from `docs/v07/PREREGISTRATION.md`, committed before any holdout bank had been played and before the sealed adversary half had been decoded. The verdicts are §13 and §15.

The primary claim

FROZEN beats `F-cheap` on both primary holdout banks, pooled over 48,000 games.

Verdict	Condition
CERTIFIED advancement	pooled lower bound above 1.53, sign replicates on both banks, gate passes
measured advancement	pooled lower bound above 0 but under 1.53, sign replicates, gate passes
not an advancement	the pooled interval contains zero, or the sign does not replicate, or the gate fails

Result: **CERTIFIED advancement**.

The secondary claims

#	Claim	Passes if	Result
S1	no collapse under partner change	at least five of the eight changed-partner deltas positive, and the minimum not below -1.00	PASS
S2	not a private convention	the off-diagonal is within 1.50 points of the diagonal	PASS
S3	no fresh adversary exploits it	every arm’s upper bound below its floor	PASS
S4	the gain is attributable	components sub-additive, no single drop exceeding the whole	PASS
S5	it survives the rule dialect	every row keeps its sign, none more than 2.00 points below default	PASS
S6	the worst case is not catastrophic	FROZEN’s worst panel cell no worse than INCUMBENT’s	PASS
S7	the instrument is intact	the planted edges recover, the sub-floor rung does not, the cheats fail exactly their own tests	PASS

The protocol’s “eight changed-partner rows” counts the self-play row among them; §13.4 reports 6 of the 7 genuinely changed rows. Both readings pass, and the phase-5 recording carries the ambiguity as deviation D9.

A failure of S1, S2 or S6 would have been a substantive failure and the report would say the release is brittle in that named way. A failure of S7 would have been procedural and the evaluation would have stopped.

What would mean v0.7 is not an advancement

The seven conditions are answered in §15.4. Two are met or unresolved. The protocol’s own closing provision is quoted in full, because this paper relies on it:

“And the honest all-negative path is a legitimate deliverable. If phase 5 returns ‘no advancement over the frontier’, the v0.7 report is a barrier report: the four ledger entries phase 3 closed with measurements, the two corrections to the corpus’s own record, the mechanical side-channel gate with its three calibrated positive controls, and the S6 residual this document commits. That is a real contribution and the report should not apologise for it.”

D The full panel

Every cell of the shared panel, four arms against 31 members, pooled over both primary banks. It is printed in full so that a reader can take the worst cell themselves rather than take §13.1’s word for which it is, and so that the minimax regret of Table 5 can be recomputed by hand.

Class **near** is 6,000 deals a bank, **far** 1,500, **expensive** 1,200. A positive number means the arm beats that member.

Panel member	Class	Frozen	Incumbent	F-cheap	Composite
F-fast	near	+4.55	+0.00	+1.83	+4.17
F-cheap	near	+3.38	−1.83	+0.00	+2.37
v05	near	+5.18	+0.69	+1.84	+4.45
v04	near	+4.66	+0.99	+2.09	+3.86
feint	near	+6.07	+3.32	+4.61	+4.93
SEALED:R-v04	near	+4.66	+0.99	+2.09	+3.86
SEALED:S-archetype-0	near	+6.07	+3.32	+4.62	+4.93
SEALED:S-ask-1	near	+4.62	−0.53	+1.01	+3.29
SEALED:S-ask-3	near	+4.08	−0.22	+1.55	+3.85
SEALED:S-decl-0	near	+4.06	−0.72	+1.10	+3.68
SEALED:S-invert-0	near	+3.93	−1.74	+0.90	+1.90
SEALED:S-reference-0	near	+4.55	+0.00	+1.83	+4.17
SEALED:S-search-1	near	+3.43	−1.81	+0.07	+2.97
SEALED:X01	near	+3.38	−0.75	+1.64	+3.35
SEALED:X01xC3f	near	+1.43	−4.24	−1.25	+1.23
SEALED:X02	near	+3.60	−1.37	+1.17	+3.04
SEALED:X13	near	+4.84	−0.30	+1.82	+4.12
SEALED:X18	near	+3.37	−0.46	+1.75	+3.22
SEALED:X20	near	+13.36	+11.62	+13.54	+13.40
v03	far	+28.72	+25.47	+24.72	+28.40
v02	far	+34.32	+31.68	+32.67	+33.17
lockout	far	+30.78	+28.65	+28.97	+29.07
detective	far	+29.65	+28.08	+27.98	+28.32
hunter	far	+46.93	+48.58	+48.65	+47.10

Panel member	Class	Frozen	Incumbent	F-cheap	Composite
diversifier	far	+43.40	+44.90	+45.18	+43.32
bluffer	far	+49.87	+49.98	+49.95	+49.88
silent	far	+32.75	+34.23	+34.05	+30.92
withholder	far	+25.35	+29.88	+29.55	+25.80
random	far	+50.00	+50.00	+50.00	+50.00
F-mid	expensive	+2.15	-3.15	-0.10	+2.00
composite	expensive	-0.04	-3.69	-2.08	+0.00

E Deviations, additions and corrections

The protocol requires a deviation for anything the evaluation did that it does not specify, recorded next to the affected result rather than in a footnote. There are 26. This register is read out of `docs/v07/FINAL-RESULTS.md` by the generator rather than retyped, so it cannot drift from the results document it reports; long entries are elided with an ellipsis and the source document carries them in full.

ID	Kind	Deviation, addition or correction
D1	Added Cell	B5 REF-FROZEN. The eleven cells B5 names cannot yield a leave-one-out DROP: the drop is FROZEN's edge minus the variant's, and FROZEN's own edge on the lattice banks is not one of the eleven. A twelfth cell was added at the lattice size on both lattice banks. B5 therefore cost 288,000 games rather than the preregistered 264,000.
D2	Added Cells	B9 planted cost and responder-vs-unhandicapped. B9.2 requires that "the recovered size tracks the planted size", which needs the planted size; B9.3 speaks of the planted excess "resolving TO zero", which is the responder's edge against the handicapped target MINUS its edge against the unhandicapped one. Four rungs x 2 banks x 6,000 deals were added for each, 192,000 games in total.
D3	Unspecified Bank	B8. Section 2.1 assigns 7090003 the role "the ablation lattice and the dialect table" but names no replicate for the dialect rows. 7090001 was used, by symmetry with B5, which the protocol does specify as 7090003 primary and 7090001 replicate.
D4	Unspecified Bank	B3 sealed rows. Section 2.1 gives 7091001 the role "evaluation bank for the sealed adversary half", but section 4/B3 puts every panel member on the same two banks as every other arm, and section 3 makes 7090001/7090002 the primary and replicate for every claim. The sealed rows were run on 7090001/7090002 with the rest of the panel, so the whole panel is scored on one shared pair of banks, as minimax regret over a shared panel requires. 7091001 was...
D5	Protocol Note	what 'tune -seed=<bank> -shard=s/4' actually plays. Section 4/B4 describes the eight searches as drawing "480 deal indices of the bank". They do not. 'tune' uses -seed only as the CEM root: it derives a fresh per-generation seed (tuner.hpp:310) and from that a per-opponent match seed (tuner.hpp:258), so the deals are a derived stream and the named bank's own deals are never constructed. Banks 7090004, 7090005 and 7091002 are therefore...

ID	Kind	Deviation, addition or correction
D6	Protocol Note	the seal is not enforced on every command. Section 2.1(b) records that ‘pathology’, ‘v7bits’ and ‘v6probe’ bypass ‘seedUsable’. ‘tune’ and ‘ablate’ also bypass it, for the same reason as D5: the seed ‘runMatch’ sees is derived, not the registered one. Phase 5 set FISH_UNSEAL_PHASE=5 once and used the seven registered seeds and no others.
D8	Added Check	reproducibility, before the material was spent. ‘runMatch’ schedules deals by work-stealing, so the deal-to-thread assignment is not fixed run to run even at a fixed thread count, and the frozen configuration carries the cross-deal agent residue of section 5.3. A scored cell was checked to be bit-identical over three runs at 13 threads and across 1, 2 and 13 threads, on training material, before any holdout bank was played.
D9	Reading	S1’s row count. Section 5.2 says "five of the eight changed-partner rows"; B6 has seven changed settings plus self-play, and the protocol’s own worked table counts eight including self-play. Both counts are printed and the verdict states which reading it used.
D10	Reading	S3’s pass and fail conditions are not complements. "Every arm’s upper bound below 1.53" and "any arm clears 1.53 with a replicated sign" leave a band in which neither holds. Both are computed and reported; if the band is occupied the verdict is reported as UNDETERMINED rather than resolved by choosing a reading.
D11	Reading	the 2.13 declaration-family floor. Section 3 correction 4 applies it to "B4’s Z03, and the declaration-accuracy column of B2". Z03 is read against 2.13. B2 has no declaration-accuracy column with an interval – ‘match –json’ emits declaration accuracy as a bare rate – so the accuracy is printed as a diagnostic and PHASE 5 MAKES NO CLAIM ABOUT THE DECLARATION CHANNEL, which is what discharges the rule.
D12	Arithmetic	S4’s naive sum. The freeze carries five components; ‘m2=0’ was dropped from it. The sum S4 tests is over those five. The six-term sum including A-m2 is printed beside it and labelled, because B5 preregisters the A-m2 cell even though the freeze does not carry the key.
D13	Note	every delta in this document is two independent intervals combined in quadrature, not a paired difference. The harness gives a paired delta only within one ‘ablate’ invocation over a shared panel, and no cell of this battery is of that form. Section 5.2 fixes this arithmetic and calls it conservative.
D14	Note	‘ablate’ was not used anywhere. It derives its own per-opponent seeds by ‘mixSeed(seed, i*7919+3)’ (main.cpp:589), so ‘ablate –seed=7090003’ would not play bank 7090003. Every lattice cell is a ‘match’ cell on the named bank instead.
D15	Note	the panel of 31 NAMED members contains three policies twice: R-v04 is v04, S-archetype-0 is feint, S-reference-0 is v06. The duplicated cells are run as named and used as a determinism check.
D16	Note	Z08 fits 55 coordinates of which 18 are inert. The protocol specifies –base=v07i for the white-box class; ‘v07i’ derives from V06Agent and reads only the 37 v0.6 coordinates (factory.hpp), while the CEM extends the vector to 55 for any v07* base. The search is over 37 live coordinates plus the inverter the base fixes.

ID	Kind	Deviation, addition or correction
D17	Note	thread counts. Every scored cell ran at <code>-threads=13</code> , which section 5.3(2) fixes for the preregistered batteries. The S6 gate condition ran at <code>-threads=1 -freshagents</code> , and the S6 residual column at <code>-threads=1</code> alone.
D18	Correction To D8	a scored cell is NOT bit-reproducible for a searching arm at panel sizes. The B0 check found five runs of a 400-deal cell bit-identical. The panel-duplicate check, on 1,500- and 6,000-deal cells, found one pair of identical commands disagreeing by one game in 12,000 – the cross-deal agent residue of section 5.3 reaching the scored mode. The magnitude is 0.008 points against a 0.63-point half-width. Reported in full under "Panel duplicates" above.
D19	Correction To D7	B0.3's changed-source NOTE could not fire, and reporting zero as "conservative" was wrong. <code>'freeze_config_v07.py -verify-only'</code> reads its drift baseline out of <code>engine/fishbot_v07.json</code> 's own <code>'provenance.srcSha256_16'</code> . That artifact was REWRITTEN after the freeze commit 0fa4a5f – at d1c6b35 and again at d8c554b – and each rewrite refreshes the baseline to the then-current tree. B0.3 therefore compared the tree against a baseline taken from the...
D20	Disclosure	five commits landed in the repository DURING the battery, and four engine sources now differ even from the refreshed baseline: <code>fish.hpp</code> , <code>game.hpp</code> , <code>human.hpp</code> , <code>table.hpp</code> , from the "web play" commits b8cb227, 2e829c2, 46f3515, 72fa936 and e0da8fd. They touch the interactive browser table, not the policy. The decisive fact is that <code>engine/fish7</code> was NEVER REBUILT: sha256 cf6d5ea2c1f0e9e3896b..., mtime 2026-08-25 13:19, identical at the start of B0 and at the...
D21	Ordering	B0 was performed before anything else, but its ARTIFACT was written later. The five B0 checks were run interactively at the top of the session, before the commit gate and before any holdout deal was played. <code>engine/p5_b0.py</code> , which re-runs all five and records them as <code>P5-B0.json</code> , was written and run about fifteen minutes into B2. Every B0 check is deterministic and reproduced identically, so the artifact records what was checked first – but its file...
D22	Observation	one action-limit hit in 428 scored cells: the phase-2 composite against SEALED:S-ask-1 on bank 7090001, one game in 12,000 (<code>limitHitRate 8.33e-05</code>). G4 gates the mirror pathology run and not scored cells, so this is not a gate violation, but the corpus standard is zero and it is recorded. NOTE that scored cells carry NO audit evidence: <code>'auditChecks'</code> is 0 in all 428 of them, because <code>'match'</code> runs the audit only under <code>'-audit'</code> and no preregistered cell...
D23	Correction	the throughput columns were printed with <code>median_high ('v[len(v)//2']</code> on an even-length list) rather than the median. Corrected to the true median; every figure moves by at most 0.11x and every one still exceeds the 3.2x the protocol records.
D25	Reading	section 6 item 3 read per bank. The protocol states the one-third location rule without a per-bank clause, and section 3's replication rule is written for "a claim whose SIGN does not replicate". Here the sign of the drop does not change between banks; the VERDICT LABEL does – LOCATED on 7090001, NOT LOCATED on 7090003. Applying section 3's rule to a threshold crossing rather than to a sign is an extension, and section 7 requires a deviation for "a...

ID	Kind	Deviation, addition or correction
D26	Substitution	B11's third clause is not computable as written. It asks phase 5 to confirm "that the holdout estimate is not systematically smaller than the training estimate by about that amount", where the estimate B11 is about is the measured holdout gain – but the protocol supplies no training figure for B2.1 or B2.2. Phase 5 substituted the three quantities the protocol DOES state in advance (the B2.4 group at +0.78, the B6 self-play row at +2.94, and B7's...
D27	Added Artifact	research/v07/results/P5-drift.json. D19's executed replacement for the vacuous hash-based drift check was performed but not initially recorded, which made it an assertion rather than evidence. It is now an artifact: the build command, both binaries' SHA-256 and byte counts, the pathology argv, the two recomputed digests, and the source sets differing from the artifact's own baseline and from the freeze commit.
D24	Correction	per-bank columns were printed unlabelled and sorted by seed NUMBER, which inverts the attribution for B5 and B8, whose primary bank is 7090003 and whose replicate is 7090001. Every per-bank column in P5-TABLES.txt is now printed as 'seed:value'. In THIS document the bank order is given in each table's first row and the pairs that follow keep it: section 5's B4 table is (7090001, 7090002) and section 9's B8 table is (7090003, 7090001). A reader...

F Reproduction and availability

Availability

The engine, every experiment artifact, the protocol, the generator that produces this paper's numbers and the provenance checker that audits them are in the project repository, <https://github.com/dylann29/fish-optimization>. The evaluation of §13 ran at commit d920f5b; the caveat immediately below is why that commit alone does not identify the run.

The run is identified by its binary, not by a commit

Five commits landed in the repository while the battery was running, so no single commit describes the tree for the whole of it. Every one of the 428 scored cells was played by `engine/fish7`, SHA-256 `cf6d5ea2c1f0e9e3896bc51f2a366834e7d60303eaf85acc553346c97359f4e3`, 1,242,312 bytes, and that binary was never rebuilt between the first bank digest and the last cell. The five later commits touch the interactive browser table and not the policy; §16.2 records the executed check — rebuilding the engine from the tree as it stood at the end of the battery and recomputing the mirror digest — that establishes the frozen policy is unmoved by them.

The battery

Where the raw output is

Under `research/v07/results/`, digested in `research/v07/results/MANIFEST-P5.json`: 29 artifacts, of which 29 re-verify against their recorded SHA-256 and byte count on disk at the time this document was generated.

Table 22: Every preregistered battery against what was measured. *Preregistered* is the count the protocol names, not the count that ran: deviations D1 and D2 add a B5 reference cell and sixteen B9 cells, because without them “no single leave-one-out drop exceeds the whole, “the recovered size tracks the planted size and “the planted excess resolves to zero are not computable from the cells the protocol lists. No cell was dropped.

Battery	What it measures	Preregistered	Added	Measured	Status
B2	headline strength against the frontier	10	0	10	complete
B3	the shared 31-member panel	248	0	248	complete
B4	a fresh adversary search	16	0	16	complete
B5	the attribution lattice	22	2	24	complete (D1)
B6	the partner-regime table	64	0	64	complete
B7	cross-play between independent runs	24	0	24	complete
B8	the rule-dialect table	16	0	16	complete
B9	the negative controls	10	16	26	complete (D2)
B10	the S6 residual	16	0	16	complete

- `P5-B0.json` — the verification block: seven bank digests, the sealed plaintext hash, the freeze round-trips, the seed registry, the engine self-audit, and the reproducibility runs.
- `P5-gate.jsonl` — the commit gate for all four configurations, rule by rule.
- `P5-B2.jsonl ... P5-B9.jsonl` — one JSON row per cell, each carrying the literal argument vector it was produced by and the complete match object the harness emitted.
- `P5-B10.jsonl` — one row per audited cell. These are seat-isolation audits rather than matches and carry a `side` object with the mismatch count and its denominator, not a match object.
- `P5-Z01.jsonl ... P5-Z08.jsonl, P5-B9-h*.jsonl` — the fitting traces: header, one record per generation, and the final weight vector each adversary spec was rebuilt from.
- `P5-TABLES.txt` — the phase-5 reduction, produced by `engine/p5\_analyse.py`.

Every row carries its own argument vector, so any single cell can be re-run by hand from the artifact without reference to this document or to the driver that produced it.

How the numbers in this document were produced

1. `engine/build\_tables\_v07.py--paper` reads the artifacts above and writes `paper/numbers\_v07\_generated.tex` and `paper/tables\_v07/*.tex`. Every macro it emits carries a comment naming the artifact and field it came from, and re-emitting a name with a different value raises rather than silently winning.
2. `paper/check\_provenance.py--versionv07` audits the result: a number used in a section must either be generated from an artifact or sit under a comment header naming an artifact that exists on disk. Of the numbers this manuscript uses, 231 are generated from the phase-5 artifacts and 151 are transcribed from the earlier phases of this programme, each attributed.
3. `paper/inline.py` flattens the modular source into a single self-contained file.

Sections 2–6 quote results from this programme’s earlier cycles. Those numbers are not retyped here either: the earlier cycles’ generators emitted them from their own artifacts, and this document loads those generated files directly. Of the numbers this paper uses, 231 are generated — most by this cycle’s generator and the remainder by an earlier one — and 151 are transcribed under a header naming the report or artifact that records them, which `paper/check\_provenance.py` verifies exists on disk.

The generated file is `\input` after the placeholder file, so a value in the placeholder can never

reach a PDF built after the battery has been run. Generated on 2026-08-26.

Sample sizes

The project's power rule is a 95% half-width of $98/\sqrt{N}$ points at parity, computed by the harness and emitted with every cell. At 48,000 games the primary claim's pooled half-width is 0.45 points; the frontier cells are sized so that the primary claim's half-width is small against the 1.53-point reference bar rather than against the effect. Far-class panel members are sized at a quarter of the near class deliberately: they are decided by 24.72 to 50.00 points and do not need the resolution.